

Farm Size, Climate Adaptation, and Unequal Climate Damages across Countries *

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Abstract

We examine how climate damages differ across countries at different stages of development through differences in farm structure and adaptation. Low-income countries are characterized by smaller, more subsistence-oriented farms and higher agricultural employment, facing greater exposure to food insecurity. Using farm-level evidence from Italy, we document that smaller farms invest substantially less in adaptation technologies and consequently experience larger climate-related productivity losses. We incorporate this evidence into a structural transformation model with heterogeneous farm sizes, climate-induced declines in agricultural productivity, and endogenous adaptation investment. We calibrate the model to Italian farm-level and aggregate data and use it to quantify the macroeconomic implications of climate damages across countries. Under plausible climate and growth scenarios, limited adaptation investment slows the reallocation of labor out of agriculture in developing countries, delaying structural transformation. This mechanism also slows convergence in aggregate labor productivity and under a less optimistic growth scenario can lead to widening cross-country income gaps.

Keywords: Climate change, productivity, agriculture, farm size, adaptation technology, structural transformation, food insecurity.

JEL classification: E14, O11, O13, O14, O4, O5, Q1, Q54.

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1 Introduction

Climate change is projected to reduce agricultural productivity in many places of the world. As agriculture is the main source of income and production for a large share of population in low-income countries, these productivity losses can have substantial aggregate consequences. Lower agricultural productivity raises food insecurity and slows the reallocation of labor out of agriculture, depressing economy-wide income and potentially widening cross-country differences in living standards (Gollin et al., 2002; Restuccia et al., 2008; Herrendorf et al., 2014). In contrast, in rich economies where the agriculture sector is small and farmers can mitigate the impact of climate change, similar climate damages have limited aggregate effects.

A key reason why a similar climate shock can have very different consequences across countries is the response through climate adaptation. Farms can mitigate climate damages by changing inputs and practices—for example through irrigation, improved varieties, and management—but adaptation is costly and uneven (Deschênes and Greenstone, 2007; Burke and Emerick, 2016). We examine how adaptation, shaped by the structure of agricultural production, mediates the aggregate impact of climate change across countries at different levels of development. We focus on farm size as a structural determinant of adaptation capacity. Agriculture in lower income countries is dominated predominantly by small farms (Adamopoulos and Restuccia, 2014); if small farms adapt less and suffer larger realized losses from climate change, then developing countries would face larger losses on agricultural productivity and more food insecurity, with substantial general-equilibrium effects. This mechanism complements recent work emphasizing limits to adaptation through sectoral reallocation in the presence of subsistence needs (Gollin et al., 2007; Nath, 2025) and general-equilibrium channels operating through trade and comparative advantage (Costinot et al., 2016).

We first document empirical facts consistent with this mechanism. Across countries, low-income economies rely more on small, subsistence-oriented farms, employ much larger shares

of workers in agriculture, and have a larger share of the population facing food insecurity. Developing countries also feature lower adaptive capacity indexes. We then use micro, farm-level, data from Italy to show that smaller farms use adaptation technologies at much lower rates and experience larger productivity losses from climate shocks. These patterns suggest that the farm-size distribution is a key determinant of the realized productivity impact of climate change.

Motivated by these facts, we develop a model of structural transformation with heterogeneous farm sizes, climate damages, and endogenous investment in adaptation technology. The model builds on [Gollin et al. \(2002\)](#) and [Restuccia et al. \(2008\)](#) and incorporates farm-size heterogeneity as in [Adamopoulos and Restuccia \(2014\)](#). Climate change reduces farm productivity, and farms choose adaptation to mitigate damages subject to a convex cost of technology investment. The model parameters are disciplined by micro estimates and calibrated to match data moments of the Italian economy.

We use the calibrated model to quantify the aggregate consequences of climate damages across economies at different levels of development. The experiments deliver two main results. First, for a common agricultural productivity shock, poorer economies adapt substantially less, leading to larger realized agricultural productivity losses, larger reallocations of labor into agriculture, larger declines in aggregate labor productivity, and large increases in the population facing food insecurity. Under a common growth scenario, these results imply a marked slowdown in the process of structural transformation in low productivity economies, and a dampening of convergence in aggregate productivity. Second, limited adaptation by small farms is a key mechanism leading to differential impacts of climate damages across countries. Limited adaptation also accounts for a sizable share of the rich–poor gap in food-security impacts.

Our paper relates to several strands of literature. First, it connects to empirical work on climate and agriculture that estimates how temperature shocks affect yields, revenues, and farm

performance (Deschênes and Greenstone, 2007; Burke and Emerick, 2016). We contribute by documenting systematic heterogeneity across farms—smaller farms adapt less and suffer larger productivity losses—and by quantifying how these micro differences translate into aggregate outcomes in general equilibrium. Second, our paper builds on the macro-development literature on cross-country income differences and structural transformation out of agriculture (Gollin et al., 2002; Restuccia et al., 2008; Herrendorf et al., 2014; Adamopoulos and Restuccia, 2014). We introduce climate damages and endogenous adaptation as forces that shape agricultural productivity, the process of structural change, and the distribution of food consumption. Third, our analysis complements recent general-equilibrium approaches to climate impacts that emphasize trade, geography, and reallocation (Costinot et al., 2016; Nath, 2025). Our paper focuses on a different adaptation angle: how the farm-size distribution shapes investment in adaptation technologies and, in turn, realized agricultural damages and food insecurity across levels of development.

The remainder of the paper is organized as follows. Section 2 describes the different datasets used in our analysis. Section 3 summarizes key cross-country empirical patterns on farm size, adaptation, and climate sensitivity, and reports reduced-form evidence from the Italian farm microdata. Section 4 presents the model. Section 5 describes the calibration to the Italian economy and reports the quantitative results from a set of experiments. Section 6 concludes.

2 Data

This section describes the datasets used in the empirical analysis. We combine (i) farm-level microdata for Italy with (ii) gridded climate data to study adaptation and productivity responses at the farm level, and (iii) global datasets on potential yields and development outcomes to document cross-country patterns in farm structure and climate exposure.

Italian farm microdata (RICA/FADN). Italy is well suited for studying farm-size heterogeneity and climate adaptation because its agricultural sector exhibits substantial dispersion in farm size, with the coexistence of very small farms and large commercial operations. This variation allows us to examine how technology adoption and climate vulnerability differ across the size distribution. Descriptive evidence on the distribution of farms, land shares, and productivity across size classes is reported in Table 1.

Our main micro dataset is the Italian Farm Accountancy Data Network (RICA), the Italian component of the EU Farm Accountancy Data Network (FADN/FSDN), administered by CREA ¹. RICA is an annual stratified survey designed to be representative of the population of commercial farms in Italy. The sampling design, with a rotating panel structure, with roughly 25 percent of farms replaced each year, follows the standard FADN methodology and stratifies farms by region, economic size, and production specialization to ensure representativeness across the agricultural sector. Each year the survey covers approximately 11,000 farms. We use the 2008–2021 waves and restrict the sample to crop-producing farms with positive operated land and non-missing information on key production variables, which are the most relevant for studying heat exposure and irrigation-based adaptation. After cleaning and applying standard sample restrictions, the final estimation sample contains approximately 85,000 farm-year observations.

The dataset contains detailed information, see Table A.1, on production, input use, land operated (hectares), labor, capital, and financial accounts. We construct a revenue-based measure of farm productivity following standard approaches in the farm productivity literature (Chen et al., 2023)². Farm size is measured as operated agricultural land in hectares, and baseline size is defined using the first year the farm is observed in the sample, ensuring that it is predetermined relative to subsequent shocks.

¹The Italian Farm Accountancy Data Network (RICA) is part of the EU Farm Accountancy Data Network (FADN). See <https://rica.crea.gov.it> for institutional details.

²Variables were constructed following Adamopoulos and Restuccia (2022), see Online Appendix.

To capture adaptation, we use the intensity of drip (micro-) irrigation adoption, measured as $\log(\text{Drip}_{it} + 1)$, where Drip_{it} denotes the area equipped with water-saving irrigation technologies. Drip irrigation is widely used as a proxy for climate adaptation because it directly improves water-use efficiency, mitigates heat and drought stress, and represents an investment decision that increases resilience to temperature shocks. Variation in this measure therefore captures differences in farms’ capacity to adopt climate-mitigating technologies across the size distribution.

Table 1: Summary Statistics by Farm Size

Farm size (ha)	Share of farms (%)	Share of land (%)	Value Added per land	Value added per worker	Capital–land ratio
< 1	2.6	0.1	57.396	1.000	16.082
1–5	16.7	1.7	12.337	1.104	5.032
5–10	18.8	4.6	6.976	1.303	3.757
10–20	20.7	9.8	4.492	1.432	2.767
20–50	24.1	25.3	2.601	1.566	2.180
> 50	17.1	58.6	1.000	1.609	1.000

Notes: Farm size bins are defined based on operated land area in hectares. Farm share and land share denote the proportion of farms and total land in each size class. Value added per hectare and the capital–land ratio are normalized relative to the values observed in the largest farm-size class. Value added per worker is normalized relative to the smallest size class. Capital refers to agricultural capital.

Climate data (E-OBS). We measure local climate conditions using the E-OBS gridded observational dataset provided by the Copernicus Climate Change Service, which reports daily weather variables across Europe at a spatial resolution of $0.1^\circ \times 0.1^\circ$ (approximately 11 km) (Cornes et al., 2018). We use daily mean temperature over the period 2008–2021, corresponding to the time window covered by the farm-level data.³

Following the climate-economics literature, we construct annual measures of temperature exposure that capture nonlinear heat stress relevant for agricultural production (e.g., Caggese

³The E-OBS dataset is publicly available from the Copernicus Climate Data Store: <https://cds.climate.copernicus.eu>. Province boundaries are obtained from official ISTAT shapefiles. Province centroids are computed using geographic overlays and matched to the nearest E-OBS grid cell.

et al., 2025). Specifically, for each grid cell and year we compute the number of hot days, defined as days with mean temperature above 30°C, and the number of very hot days, defined as days with mean temperature above 35°C. These measures are obtained by aggregating daily observations within each year. To match climate conditions to farms, we exploit the geographic location of farms at the province level. Climate variables are aggregated at the province-year level and merged with the farm panel according to the farm’s province and year of observation. This approach balances spatial precision with data availability and is commonly used when exact farm coordinates are not observed.

Italy exhibits substantial climatic heterogeneity due to its latitudinal extension, complex topography, and strong regional variation in temperature patterns. Southern regions and inland areas experience substantially higher exposure to extreme heat compared to northern regions, providing meaningful cross-sectional variation for identifying the impact of temperature shocks on farm outcomes.

Potential yields (GAEZ). For global and cross-sectional analyses, we use potential yield estimates from the Global Agro-Ecological Zones (GAEZ) database produced by FAO and IIASA. GAEZ provides high-resolution predictions of potential yields (tons per hectare) under given agro-climatic conditions, based on agronomic models that incorporate soil, elevation, and climate. We focus on baseline potential yields around the year 2000 (e.g., for wheat as a benchmark crop) and aggregate grid-cell outcomes to provinces or countries when needed. These measures provide a proxy for underlying agricultural productivity and a disciplined way to quantify climate-induced changes in potential yields.

Cross-country development and farm structure. To relate farm structure to development, we combine national measures of income from the Penn World Tables (GDP per capita in PPP terms) with historical series from the Maddison Project Database. We use country-level measures of average farm size (hectares), constructed from agricultural census

and survey sources (e.g., FAO/WCA), and merge these with the development indicators. These data are used to document stylized facts on how farm size, agricultural employment, and exposure to climate damages vary across the income distribution.

Merging and construction. For Italy, we merge RICA to E-OBS at the province-year level and then estimate farm-level relationships between productivity/adaptation outcomes and temperature exposure. For the global component, we align GAEZ potential yields and climate measures to a common grid and then aggregate to the country level before merging with income and farm-size indicators. Throughout, we ensure consistent spatial and temporal alignment across sources and verify that results are robust to alternative assignment rules (nearest-neighbor vs. within-area averaging) and alternative climate summaries.

3 Empirical Findings

We document a set of empirical findings that motivate the main mechanisms in the model. We first provide cross-country evidence that global warming is projected to decrease potential yields in agriculture, an effect that differs substantially across countries but not systematically with respect to their level of labor productivity. We also document that more people are engaged in agriculture in less developed countries, operate farms at a smaller average land scale, and feature lower adaptation capacity to climate change. We then provide farm-level evidence that climate shocks affect disproportionately the productivity of smaller farms because they invest less in adaptation. In particular, the evidence shows that adaptation is size-dependent: smaller farms invest less in mitigation technologies and therefore experience larger productivity losses from heat shocks. Heat-related productivity losses are steeper at smaller farm sizes, and investment in adaptation attenuates the heat–productivity relationship. This is consistent with the cross-country evidence on adaptive capacity.

3.1 Cross-Country Evidence

Fact 1: Global warming reduces potential yields. Global temperatures have risen substantially since the pre-industrial period and are projected to increase further over the twenty-first century. Because crop growth is highly sensitive to temperature and water stress, continued warming implies sizable reductions in potential agricultural productivity. We highlight three patterns. First, global mean temperature has increased and continues to rise under alternative emissions trajectories (Figure 1). Second, warming is uneven across space, so the increase in heat exposure differs sharply across regions (Figure 2). Third, these temperature changes translate into large and spatially heterogeneous declines in potential yields of major staple crops, with the largest losses concentrated in already warm regions (Figure 3).

Table 2: Yield Response to a 1°C Temperature Increase

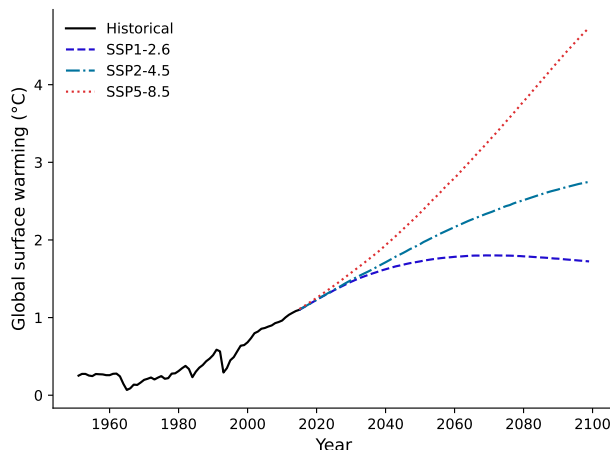
Crop	Yield change (%)	Source
Maize	−7.4	Zhao et al. (2017)
Wheat	−6.0	Zhao et al. (2017)
Rice	−3.2	Zhao et al. (2017)
Soybean	−3.1	Zhao et al. (2017)
Wheat (multi-model mean)	−6.0 to −8.5	Liu et al. (2019)
Global cereals (range)	−3.0 to −7.0	Rosenzweig et al. (2014) ; IPCC (2021)

A large empirical and simulation-based literature finds sizable negative effects of higher temperatures on crop yields. Across meta-analyses and crop-model ensembles, a 1°C increase in growing-season temperature reduces yields of major cereals by roughly 3–7% on average. For example, [Zhao et al. \(2017\)](#) estimate yield elasticities of −7.4% for maize, −6.0% for wheat, −3.2% for rice, and −3.1% for soybean per 1°C. These magnitudes are consistent with the projected yield losses in Figure 3, and with the broader conclusion that warming generates substantial risks for agricultural productivity, especially in warm regions.

Figure 1 plots observed global temperature anomalies together with projections from the CMIP6 multi-model ensemble. Relative to the 1850–1900 baseline, global mean temperature

has already increased by about 1.1°C . Under a low-emissions pathway (SSP1–2.6), warming stabilizes around 2°C by 2100; under a middle pathway (SSP2–4.5), it reaches roughly 3°C ; and under a high-emissions pathway (SSP5–8.5), it exceeds 5°C . The key point is that further warming is projected across scenarios, with large differences in magnitude depending on emissions.

Figure 1: Global Average Temperature Projections (CMIP6)

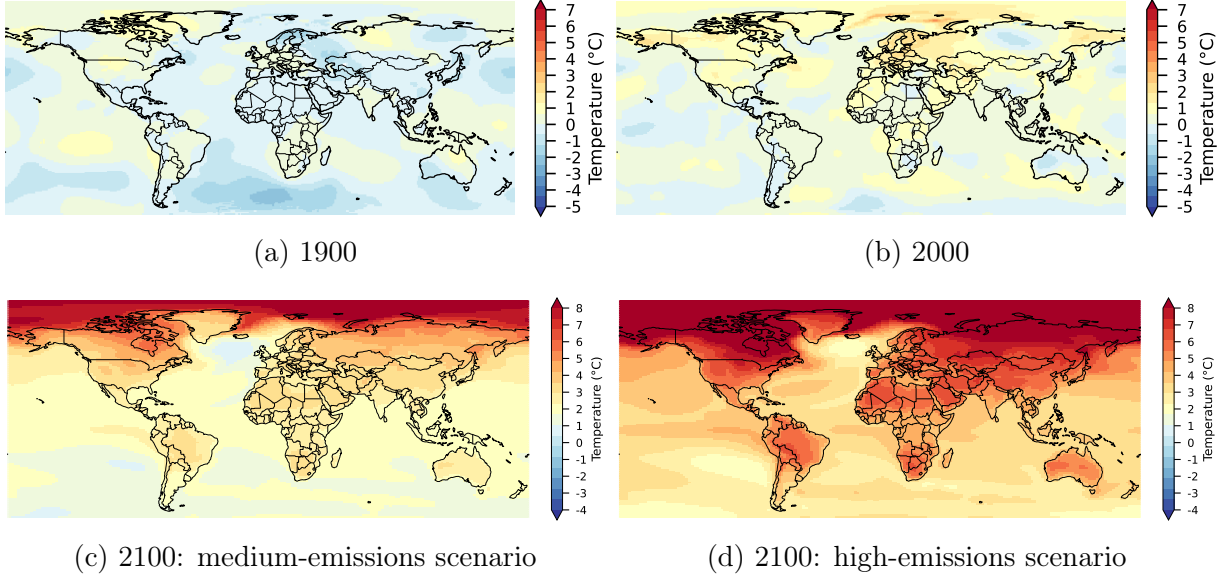


Notes: Temperature anomalies are relative to the 1850–1900 average. Data are from [Fyfe et al. \(2022\)](#). The solid black line shows historical observations. The dashed blue, dash-dotted navy, and dotted red lines show projections under SSP1–2.6, SSP2–4.5, and SSP5–8.5, respectively.

Figure 2 shows that the rise in temperature is not uniform. Some regions experience substantially larger increases in heat exposure than others, reflecting both geography and baseline climate conditions. These spatial differences matter for agriculture because yield responses are highly nonlinear: damages rise sharply when temperatures move beyond crop-specific thresholds. The unequal pattern of warming therefore implies unequal pressure on attainable productivity across regions.

Figure 3 summarizes projected changes in potential yields the 2050s (period 2041–2070) from two climate models, GFLD-ESM2M (climate model “M1” below) and IPSL-CM5A-LR (climate model “M2” below), with CO₂ fertilization and with only rainfed water supply. For each country, we construct a crop-share-weighted average potential yield across major staples, and plot the ratio of future to current potential yields. Values below one indicate

Figure 2: Temperature Anomaly ($^{\circ}\text{C}$)



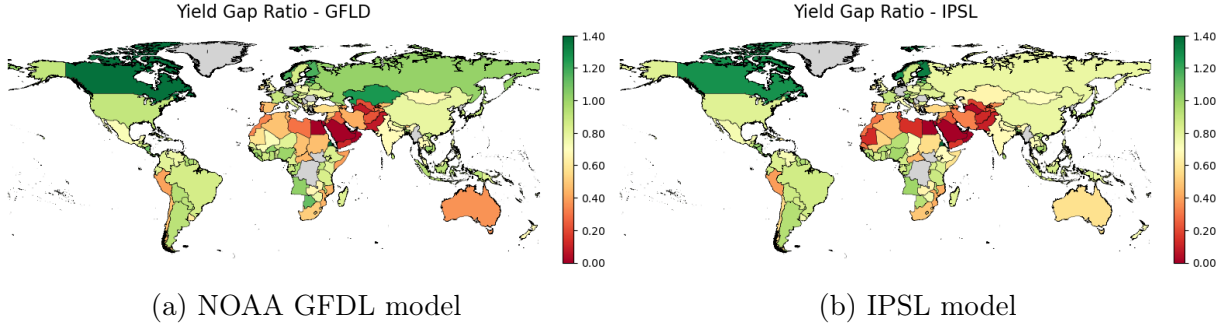
Notes: Temperature anomalies are relative to the 1850–1900 average. Data are from [Rohde and Hausfather \(2020\)](#). Panels show historical anomalies (1900, 2000) and projected anomalies for 2100 under medium- and high-emissions scenarios.

projected declines in attainable yields under otherwise favorable production conditions. The projections imply large and uneven losses. Yield declines are concentrated in low-latitude, already warm regions, while some high-latitude regions exhibit modest gains.

Figure 4 plots the ratio of projected potential yields in the 2050s (2041–2070) relative to the 1981–2010 baseline against labor productivity relative to Italy. Across both climate models (NOAA GFDL and IPSL), the figure shows no strong systematic relationship between projected potential-yield changes and country income level. In other words, climate-driven yield impacts are not concentrated only among poorer or richer countries when viewed through this cross-country income gradient.

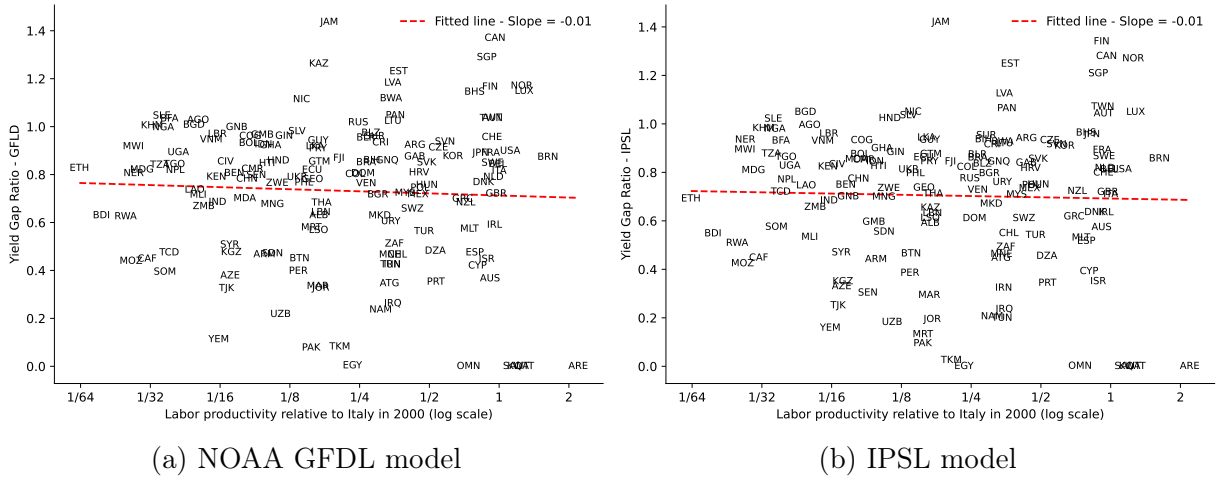
The figure indicates substantial average losses and large dispersion within income groups. On average, countries face projected potential-yield declines of about 30% (ratios around 0.7), but this mean effect masks pronounced heterogeneity among countries with similar initial productivity levels. In particular, many low- and middle-income countries experience much

Figure 3: Projected Change in Potential Yields under a High-Emissions Scenario



Notes: Data are from the Global Agro-Ecological Zones (GAEZ) v4 database. The figure reports the ratio of projected to current potential yields for each country, where potential yields are computed under high-input and high-water-availability assumptions. Projections compare 2041–2070 to 1981–2010 under a high-emissions scenario. Country-level outcomes are constructed as crop-share-weighted averages across major staple crops, using baseline crop shares. The two panels use climate projections from the NOAA *Geophysical Fluid Dynamics Laboratory* (GFDL) model and the *Institut Pierre-Simon Laplace* (IPSL) model. Values below one indicate yield losses relative to current potential yields.

Figure 4: Projected Potential Yields, 2041–2070 relative to 1981–2010



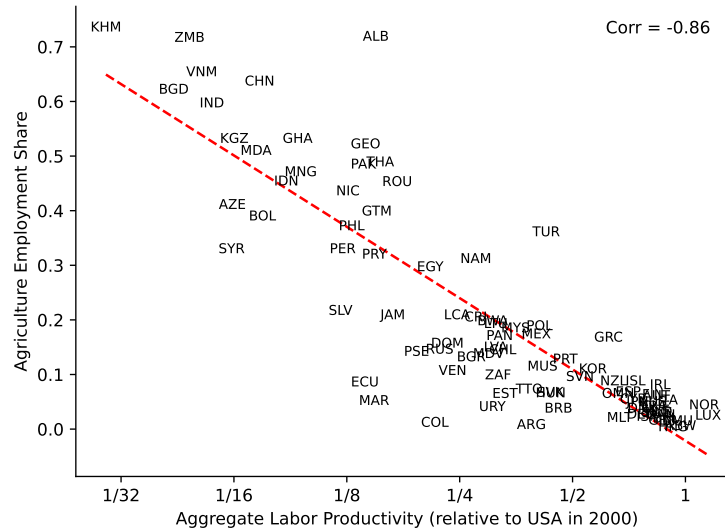
Notes: Data are from the Global Agro-Ecological Zones (GAEZ) v4 database. The figure reports the ratio of projected to current potential yields for each country, where potential yields are computed under high-input and high-water-availability assumptions. Projections compare 2041–2070 to 1981–2010 under a high-emissions scenario. Country-level outcomes are constructed as crop-share-weighted averages across major staple crops, using baseline crop shares. The two panels use climate projections from the NOAA *Geophysical Fluid Dynamics Laboratory* (GFDL) model and the *Institut Pierre-Simon Laplace* (IPSL) model. Values below one indicate yield losses relative to current potential yields.

larger projected losses, with potential yields falling by 70%–90% in the most affected cases.

Fact 2: Poorer countries rely more on agriculture and operate at smaller scale.

A second set of facts show how agricultural share varies with income. In poorer countries, a much larger share of the labor force works in agriculture, and the sector accounts for a larger fraction of aggregate production. As countries develop, labor reallocates out of agriculture toward more productive sectors ([Gollin et al., 2002](#); [Restuccia et al., 2008](#); [Gollin et al., 2014](#)). Figure 5 illustrates this pattern: agricultural employment shares are high in low-income economies and decline sharply with GDP per capita. This matters for climate vulnerability because a given fall in agricultural productivity mechanically has a larger aggregate impact when agriculture employs a larger share of workers and production.

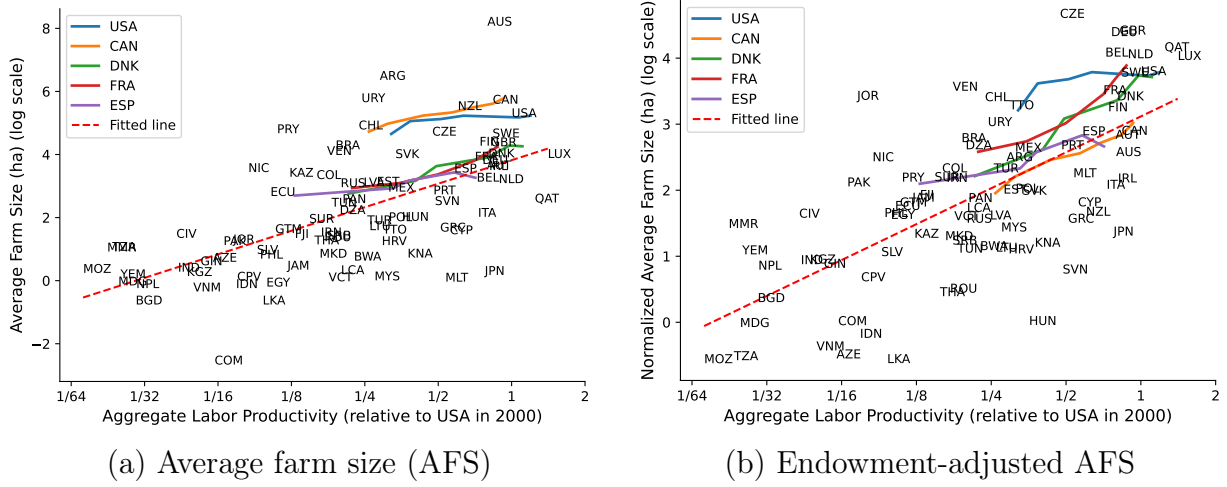
Figure 5: Share of Employment in Agriculture and GDP per Capita (2000)



Notes: Country codes denote country observations. Agricultural employment shares are from FAOSTAT. GDP per capita is from the Penn World Table ([Feenstra et al., 2015](#)).

A closely related fact is that farming is much more small-scale in poorer countries. Average farm size is strongly increasing in income (Figure 6). Panel (a) plots average farm size against GDP per capita and shows a steep positive relationship. Panel (b) adjusts farm size for cross-country differences in agricultural land endowments; the relationship remains, indicating that land availability alone does not account for the pattern. Following [Adamopoulos and](#)

Figure 6: Average Farm Size and GDP per Capita across Countries (2000)



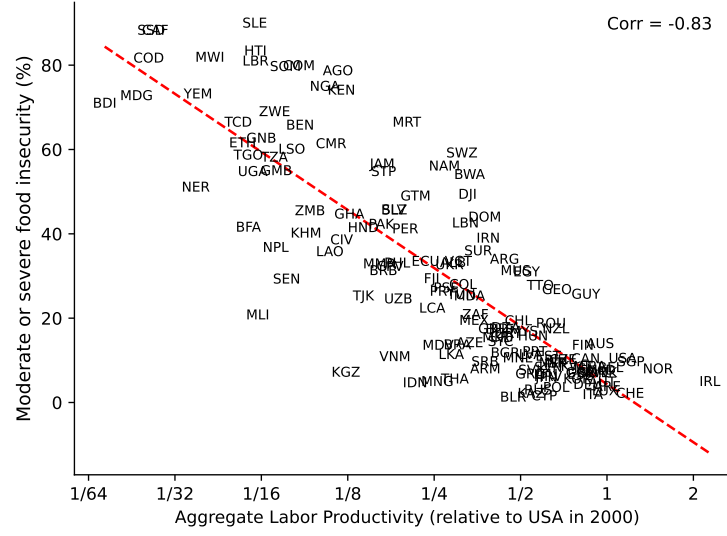
Notes: Country codes denote country observations. Farm size is measured in hectares (log scale). Panel (b) adjusts farm size for agricultural land endowments by scaling each country's land per capita to match the United States. Farm size data are from FAOSTAT. GDP per capita is from the Penn World Table (Feenstra et al., 2015).

Restuccia (2014), this points to frictions—including land-market distortions, tenure systems, and constraints on capital deepening—that limit consolidation and prevent scale from rising in poorer economies. This scale gap is central for our mechanism because many mitigation technologies involve fixed costs and complementary investments, so technology investment incentives are likely to differ by scale.

Finally, food insecurity is far more prevalent in low-income economies (Figure 7). The fraction of the population facing moderate or severe food insecurity is close to zero in most high-income countries but remains large in many poorer countries. This pattern indicates that a larger share of households is close to subsistence levels of food consumption, so climate-driven productivity declines are more likely to generate meaningful welfare losses through heightened food insecurity.

Taken together, these cross-country facts imply that poorer countries are more exposed to climate risk: agriculture is larger, farms are smaller, and more households operate close to subsistence. The next section turns to micro evidence on whether small farm size is

Figure 7: Food Insecurity and GDP per Capita



Notes: Country codes denote country observations. Food insecurity is the share of the population facing moderate or severe food insecurity (percent), from FAOSTAT. GDP per capita is from the Penn World Table (Feenstra et al., 2015).

associated with lower adaptation and larger heat-related productivity losses, and whether adaptation can mitigate these losses.

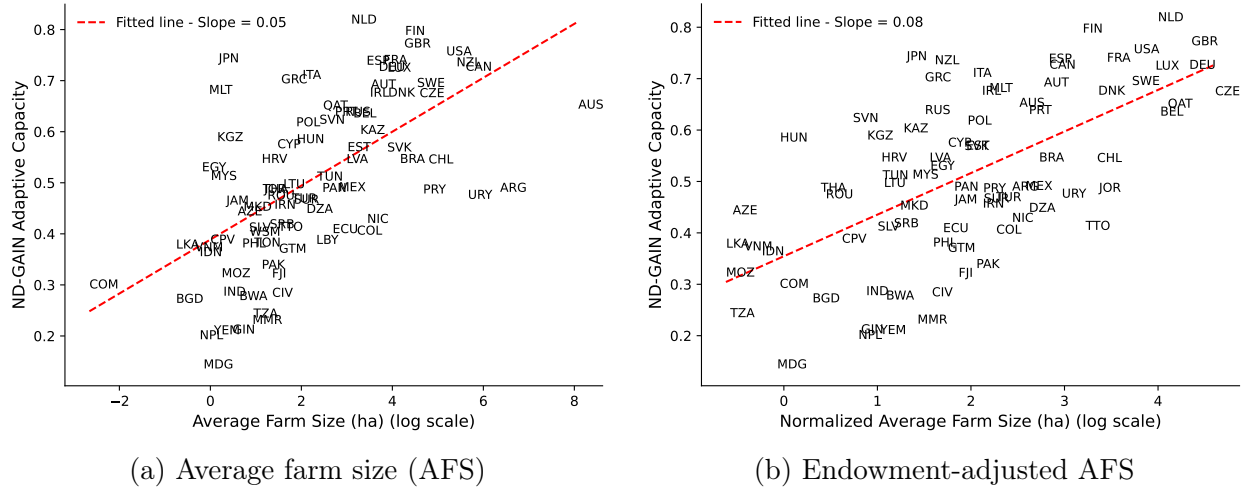
Fact 3: Countries with higher average farm size exhibit stronger climate adaption capacity. We combine measures of average farm size with a country-level index of adaptive capacity from the ND-GAIN dataset. The ND-GAIN adaptive capacity index summarizes a country’s readiness and ability to cope with climate risks through economic, institutional, and infrastructural conditions. In the data, countries with larger farms exhibit systematically higher adaptive capacity.

Figure 8a plots the ND-GAIN adaptive capacity index against average farm size. The relationship is positive and strong: countries with small farms tend to score low on adaptive capacity, while countries with large farms tend to score high. This correlation is robust to adjusting farm size for differences in agricultural land endowment. Figure 8b repeats the exercise using “endowment-adjusted” farm size, constructed by scaling each country’s average farm size to the counterfactual level it would have if the country had the same agricultural

land per capita as the United States. The positive association remains, suggesting that the farm size gradient in adaptive capacity does not simply reflect land abundance.

The cross-country data provides further evidence suggesting that small farm tend to have weaker adaptive capacity, consistent with a lower propensity to adopt mitigation technologies and a higher vulnerability to climate shocks.

Figure 8: Adaptive Capacity and Average Farm Size across Countries



Notes: Panel (a) plots the ND-GAIN adaptive capacity index against average farm size (hectares per holding). Panel (b) uses an endowment-adjusted farm size measure that scales each country's average farm size by the ratio of its agricultural land per capita to that of the United States, so that the horizontal axis reflects the implied farm size if all countries had the same agricultural land per capita as the United States. The adaptive capacity index is from the ND-GAIN Country Index; farm size and agricultural land are from FAOSTAT.

3.2 Farm-Level Evidence

Fact 4: Smaller farms are more climate-vulnerable because they invest less in adaptation. We now turn to farm-level evidence on how Heat Shocks interact with farm size and adaptation technology investment. Using microdata from the Italian farm panel (RICA), we document three patterns. First, Heat Shocks reduce productivity growth, and the losses are systematically larger for smaller farms. Second, investment of adaptation technologies is increasing in farm size: smaller farms adapt less. Third, adaptation attenuates the productivity impact of Heat Shocks, with causal evidence from an instrumental-variable

strategy.

Productivity response to Heat Shocks and baseline farm size. We begin by estimating how Heat Shocks affect farm productivity growth and whether the effect depends on baseline size:

$$\Delta \log(\text{TFP}_{it}) = \beta_0 \log(\text{FarmSize}_{i0}) + \beta_1 \text{HeatShock}_{it} \quad (1)$$

$$+ \beta_2 (\text{HeatShock}_{it} \times \log(\text{FarmSize}_{i0})) + \gamma_t + X'_{it}\theta + \varepsilon_{it}, \quad (2)$$

where $\Delta \log(\text{TFP}_{it})$ is farm-level TFP growth, and $\log(\text{FarmSize}_{i0})$ is baseline farm size measured at the start of the panel. We measure HeatShock_{it} , as shown in the columns in Table 3, as the number of days with temperature above 35°, number of days with temperature within 30°- 35° and the annual deviation of local temperature, following the standard approach in the climate–agriculture literature, γ_t controls for common shocks. Identification comes from within-farm variation in local weather over time, with baseline size predetermined relative to contemporaneous shocks.

The estimates imply $\beta_1 < 0$ and $\beta_2 > 0$: Heat Shocks reduce productivity growth on average, but the losses are smaller for larger farms. Appendix Table B.6 reports average marginal effects by baseline size quintile and shows that damages are concentrated among smaller farms, becoming much weaker for the largest farms. Appendix Figure B.1 provides complementary evidence from quantile regressions, which show that the attenuation by size is particularly pronounced at higher percentiles of the productivity distribution.

Farm size and drip irrigation. We next test whether drip irrigation is size-dependent:

$$\begin{aligned} \log(\text{Drip}_{it} + 1) = & \gamma_1 \log(\text{FarmSize}_{i0}) + \gamma_2 \text{HeatShock}_{it} \\ & + \gamma_3 \log(\text{FarmSize}_{i0}) \cdot \text{HeatShock}_{it} + \alpha_i + \gamma_t + X'_{it}\theta + u_{it} \end{aligned} \quad (3)$$

In order to assess that dependence we use $\text{Log}(\text{Drip}_{it} + 1)$, where Drip is the area irrigated

Table 3: Heat Shocks, Farm Size, and Farm TFP

<i>Dependent variable:</i> $\Delta \log(\text{TFP}_{it})$	(1)	(2)	(3)
<i>Heat Shock measure:</i>	# Days > 35°C	# Days 30–35°C	ΔTemp
$\log(\text{FarmSize}_{i0})$	0.095*** (0.019)	-0.061 (0.045)	-0.007 (0.029)
HeatShock	-0.043*** (0.008)	-0.008** (0.004)	-0.088*** (0.030)
$\log(\text{FarmSize}_{i0}) \times \text{HeatShock}$	0.012*** (0.004)	0.002** (0.001)	0.003*** (0.005)
Constant	7.101*** (0.067)	7.548*** (0.152)	7.345*** (0.105)
Observations	85,515	85,515	61,069
Year $\times$ sector FE	✓	✓	✓
Year $\times$ altitude FE	✓	✓	✓

Notes: The unit of analysis is a farm–year. The dependent variable is farm-level TFP growth, $\Delta \log(\text{TFP}_{it})$. The sample includes crop-producing farms only. Columns (1)–(3) estimate equation (1) using alternative measures of heat shocks, constructed as deviations from the location-specific long-run mean (local climate normal): the number of very hot days (> 35°C), the number of hot days (30–35°C), and the anomaly in annual temperature (ΔT). All specifications include year-by-sector and year-by-altitude fixed effects. Standard errors are clustered at the provincial level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

with micro(Drip) irrigation as a proxy of adaptation, the annual variation of temperature as heat shocks and the initial farm size. Table 4 shows that Heat Shocks reduce adaptation on average, while the positive interaction term implies that adaptation responses are weaker for larger farms and more pronounced among small farms. The same qualitative pattern holds when we condition on baseline productivity rather than baseline size (Table B.5) and when we examine marginal effects across the baseline size distribution (Appendix Figure B.2).

Adaptation mitigates productivity losses: OLS and IV evidence. Finally, we estimate whether adaptation flattens the temperature–productivity relationship:

$$\begin{aligned}
\Delta \log(\text{TFP}_{it}) = & \beta_1 \log(\text{Adapt}_{it}) + \delta_1 \text{HeatShock}_{it} \\
& + \delta_2 (\text{HeatShock}_{it} \times \log(\text{Adapt}_{it})) + X'_{it} \theta + \gamma_t + \eta_{it},
\end{aligned} \tag{4}$$

Table 4: Farm Size, Heat Shocks, and Adaptation (Drip Irrigation)

<i>Dependent variable:</i> $\log(\text{Drip}_{it} + 1)$	(1)	(2)	(3)
<i>Heat Shock measure:</i>	Δ Temperature		
$\log(\text{FarmSize}_{i0})$	0.091*** (0.002)	0.084*** (0.019)	0.085*** (0.019)
HeatShock		-0.036*** (0.012)	-0.058*** (0.015)
$\log(\text{FarmSize}_{i0}) \times \text{HeatShock}$			0.008** (0.004)
Constant	0.062*** (0.007)	0.080* (0.046)	0.079* (0.046)
Observations	85,514	61,069	61,069
Year $\times$ sector FE	✓	✓	✓
Year $\times$ altitude FE	✓	✓	✓

Notes: The unit of analysis is a farm-year. The dependent variable is $\log(\text{Drip}_{it} + 1)$. “Heat Shock” is measured as the change in average temperature relative to the previous year at the same location. All specifications include year-by-sector and year-by-altitude fixed effects. Standard errors are clustered at the provincial level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

where $\log(\text{Adapt}_{it})$ measures adaptation (e.g., irrigated area equipped with water-saving technologies) and HeatShock_{it} is the temperature anomaly defined above. The coefficient of interest is δ_2 : a positive value implies that farms with greater adaptation experience smaller productivity losses when Heat Shocks increase.

Because adaptation can be endogenous, we instrument $\log(\text{Adapt}_{it})$ and $\text{HeatShock}_{it} \times \log(\text{Adapt}_{it})$ using farm-level access to specific EU environmental and agro-climatic investment funds and the interaction of these funds with HeatShock_{it} . The first stage is strong, and 2SLS estimates imply that adaptation significantly mitigates the productivity impact of Heat Shocks among farms whose adaption responds to these funding opportunities.

The 2SLS coefficient on the interaction term is positive and statistically significant, indicating that adaptation mitigates the productivity impact of Heat Shocks. Interpreting the interaction in (4), a 1% increase in adaptation intensity reduces the marginal TFP loss from

Table 5: Adaptation, Heat Shocks, and TFP Growth: OLS and IV (2SLS) Estimates

	(1) OLS	(2) OLS	(3) 2SLS	(4) 2SLS
<i>Dependent variable: $\Delta \log(\text{TFP}_{it})$</i>				
$\log(\text{Adapt}_{it})$	0.012 (0.007)	0.018** (0.008)	0.045** (0.019)	0.057** (0.025)
$\text{HeatShock}_{it} \times \log(\text{Adapt}_{it})$	0.019 (0.023)	0.024 (0.021)	0.126** (0.062)	0.103** (0.045)
HeatShock_{it}	-0.058 (0.054)	-0.055 (0.050)	-0.184** (0.089)	-0.147* (0.076)
Observations	3,079	3,270	3,079	3,270
Year FE	✓	✓	✓	✓
Sector FE	✓	✓	✓	✓
Altitude FE	✓	✓	✓	✓
Kleibergen–Paap F			11.990	19.980

Notes: The unit of analysis is a farm-year. The dependent variable is farm-level TFP growth, $\Delta \log(\text{TFP}_{it})$. HeatShock_{it} is the temperature anomaly, defined as the deviation from the location-specific long-run mean. The endogenous regressors are $\log(\text{Adapt}_{it})$ and $\text{HeatShock}_{it} \times \log(\text{Adapt}_{it})$, instrumented using the log of EU environmental and agro-climatic funds received by the farm and its interaction with HeatShock_{it} . All specifications include year, sector, and altitude fixed effects. Standard errors are clustered at the province level.

a 1°C increase in the temperature anomaly by roughly 0.10–0.13 percentage points in our baseline specifications.

4 Model

We develop a model of structural transformation between agriculture and non-agriculture, building on [Gollin et al. \(2002\)](#) and [Restuccia et al. \(2008\)](#). The model incorporates production heterogeneity in agriculture as in [Adamopoulos and Restuccia \(2014\)](#) and introduces productivity losses from climate change. Farmers choose whether to invest in climate adaptation technologies. We examine how technological and institutional factors that shape the farm size distribution influence climate adaptation decisions and their macroeconomic consequences for structural transformation and productivity.

4.1 Economic Environment

Technologies. At each date two goods (agriculture and non-agriculture) are produced in sectors. A homogeneous agricultural good is produced by farms indexed by i , with a decreasing returns to scale production function given by

$$y_i = A\kappa(s_i z)^{1-\gamma}\ell_i^\gamma, \quad \gamma \in (0, 1),$$

where y_i denotes agricultural output, ℓ_i is land input, A represents economy-wide productivity, κ is agriculture-specific productivity, and $(s_i z)^{1-\gamma}$ captures the farm's idiosyncratic productivity which has two components. The component s_i reflects farmer's idiosyncratic ability which is drawn from an known and exogenous distribution. The component z reflects idiosyncratic productivity determined by an exogenous climate shock D and the farmer's endogenous investment choices in productivity-enhancing and climate adaptation technology x , which we describe in detail below.

The homogeneous non-agricultural good is produced by a representative firm using labor according to the following constant returns to scale production function:

$$Y_n = AN_n,$$

where A is economy-wide productivity and N_n is the labor input.

Market structure. We assume competitive markets in both sectors and normalize the price of agriculture to one (numeraire). We denote by p_n the relative price of non-agriculture, q the rental rate of land, and w the wage rate in non-agriculture.

Preferences and endowments. The economy is populated by a unit mass of individuals. Each individual owns a fixed equal share of the aggregate land stock. Total land supply is exogenous and fixed at L , and is supplied inelastically to farms for agricultural production.

Individuals have Price-Independent Generalized Linear (PIGL) preferences over agricultural and non-agricultural consumption (Boppart, 2014). Indirect utility as a function of the non-agricultural price p_n and income E is

$$v(p_n, E) = \frac{1}{1-\varphi} E^{1-\varphi} - \frac{\alpha}{1-\rho} p_n^{1-\rho} - \frac{1}{1-\varphi} + \frac{\alpha}{1-\rho},$$

where α is the preference weight on agricultural consumption, φ is the income elasticity, and ρ is the price elasticity to p_n .

The implied Hicksian demands for agricultural and non-agricultural goods are

$$c_a = \alpha E^\varphi, \quad c_n = \frac{E - c_a}{p_n} = \frac{E - \alpha E^\varphi}{p_n},$$

where c_a and c_n denote consumption of the agricultural and non-agricultural good, respectively.

Occupational choice. Following Adamopoulos and Restuccia (2014), we abstract from occupational selection by assuming that all individuals are ex-ante identical and only observe their farm operating ability after choosing an occupation. Selection has been studied extensively in the literature (Lagakos and Waugh, 2013; Adamopoulos et al., 2022) and we focus instead on the farm size distribution and heterogeneity *within* the agricultural sector. Because labor is perfectly mobile across sectors, in equilibrium, the expected value of operating a farm for an individual equals the non-agricultural wage, so that individuals are indifferent between operating a farm or working in non-agriculture for a wage. Formally, with $V(s)$ denoting the value from operating a farm for ability s and w the non-agricultural wage, an interior occupational allocation requires

$$\mathbb{E}_s[V(s)] = w. \tag{5}$$

After choosing to operate a farm, individuals farm ability is drawn from an iid exogenous distribution $s_i \sim F(s)$. Realized income for a farmer is given by the farm value $V(s_i)$, while an individual working in non-agriculture earns the wage w . In addition, all individuals receive their share of aggregate land rents qL , so that individual income is the sum of sectoral income (farm profit or wage) and land income.

4.2 Equilibrium

The model is static and we consider a competitive equilibrium in which households, farms, and firms take prices as given, and prices clear the markets.

Incumbent farms. An incumbent farm i is characterized by idiosyncratic productivity s_i and its choice of technology z . The farm's expected per-period profit is given by

$$\pi(s_i z) = \max_{\ell \geq 0} A\kappa(s_i z)^{1-\gamma} \ell^\gamma - q\ell.$$

The optimal land choice and output of an incumbent farm is given by

$$\begin{aligned} \ell(s_i z) &= \left(\frac{\gamma}{q}\right)^{\frac{1}{1-\gamma}} (A\kappa)^{\frac{1}{1-\gamma}} s_i z, \\ y(s_i z) &= \left(\frac{\gamma}{q}\right)^{\frac{\gamma}{1-\gamma}} (A\kappa)^{\frac{1}{1-\gamma}} s_i z. \end{aligned}$$

The optimal expected profit of an incumbent farm is given by

$$\pi(s_i z) = (1 - \gamma) \left(\frac{\gamma}{q}\right)^{\frac{\gamma}{1-\gamma}} (A\kappa)^{\frac{1}{1-\gamma}} s_i z.$$

Climate damage and farm technology. Each farm i is subject to an exogenous and common climate damage shock $D \geq 0$ that in isolation lowers agricultural productivity. We capture climate damage and farm technology by specifying farm technology $z_i \in (0, \infty)$ as a

function of the extent of climate damage D and farm investment in productivity-enhancing and adaptation technology (e.g. irrigation, heat-tolerant varieties, soil conservation) $x_i \geq 0$,

$$z_i = g(D, x_i),$$

where the function $g : \mathbb{R}_+^2 \rightarrow (0, \infty)$ satisfies

$$g_D(D, x) < 0, \quad g_x(D, x) > 0, \quad g_{xx}(D, x) \leq 0,$$

implying that climate damage reduces productivity and farm investment raises productivity with diminishing marginal returns.

For the quantitative analysis we parameterize farm technology as

$$z(D, x) = e^{-D} (1 + x)^{\eta_0 + \eta_1 D}, \quad \eta_0 \geq 0, \eta_1 \geq 0. \quad (6)$$

Note that when $D = 0$, η_0 controls the return to farm productivity-enhancing investment technology independent of climate.

We emphasize that the above specification (in logs) maps into the empirical regression discussed earlier:

$$\log z(D, x) = -D + (\eta_0 + \eta_1 D) \log(1 + x). \quad (7)$$

In the absence of farm investment ($x = 0$), $z(D, 0) = e^{-D}$, D captures the pure climate damage component to farm productivity. The parameter η_0 governs the return to productivity-enhancing investment independent of climate, while η_1 controls the return to adaptation investment to the level of climate damage. When $\eta_0 > 0$, farm investment raises productivity even in the absence of damage ($D = 0$) and $\eta_1 > 0$, adaptation becomes more effective in mitigating damage as D increases. The magnitude of these elasticity parameters determine the extent to which farm investment is productivity-enhancing and adaptation.

We assume a convex cost function of farm investment $k : \mathbb{R}_+ \rightarrow \mathbb{R}_+$,

$$k(x) = \frac{\psi}{1+\nu} x^{1+\nu}, \quad \psi > 0, \nu \geq 0, \quad (8)$$

where ψ is a scale parameter and ν controls the curvature.

Given the profit function $\pi(sz) = \Omega sz$ derived above, an incumbent farm with productivity s facing damage D chooses farm investment x to solve

$$\Pi(s, D, x) = \Omega s z(D, x) - k(x) = \Omega s e^{-D} (1+x)^{\eta_0+\eta_1 D} - \frac{\psi}{1+\nu} x^{1+\nu}. \quad (9)$$

The first-order condition for an interior solution $x > 0$ is

$$\frac{\partial \Pi}{\partial x} = \Omega s e^{-D} (\eta_0 + \eta_1 D) (1+x)^{\eta_0+\eta_1 D-1} - \psi x^\nu = 0. \quad (10)$$

Because (10) is nonlinear in x and involves both $(1+x)$ and x raised to powers, it does not admit a simple closed-form solution for $x^*(s, D)$. In the quantitative analysis we therefore solve (10) numerically for $x^*(s, D)$, imposing the Kuhn–Tucker conditions $x \geq 0$ and complementary slackness, and then set

$$z(s, D) \equiv z(D, x^*(s, D)), \quad V(s, D) = \Omega s z(s, D) - k(x^*(s, D)).$$

Definition of equilibrium. A competitive equilibrium comprises a set of prices (p_n, w, q) , agricultural and non-agricultural labor allocations (N_a, N_n) ; farm-level decision rules for land demand $\ell(s, D)$, productivity investment $x(s, D)$, climate-adjusted productivity $z(s, D)$, output $y(s, D)$, profit $\pi(s, D)$, and farm value $V(s, D)$; and individual consumption allocations for farmers $\{c_a^F(s), c_n^F(s)\}$ and workers (c_a^W, c_n^W) such that:

- (i) **Farm problem.** Given prices (q, w, p_n) , policy functions $x(s, D)$ and $\ell(s, D)$ solve the incumbent farm with ability s facing climate damage D , yielding the optimal per-period

profit $\pi(s, D)$ and value net of adaptation cost $V(s, D)$.

- (ii) **Occupational choice.** An interior agricultural employment share $N_a \in (0, 1)$ requires that the expected value of operating a farm equals the non-agricultural wage:

$$\mathbb{E}_s[V(s, D)] = w. \quad (11)$$

- (iii) **Non-agricultural firm problem.** Given prices (p_n, w) , N_n solves the representative non-agricultural firm's problem, which implies $p_n = w/A$ and the firm is indifferent on the level of labor demand N_n .
- (iv) **Consumption allocations.** Given farm income $I^F(s)$ and the relative price p_n , the pair $(c_a^F(s), c_n^F(s))$ solves the consumption problem of a farm operator with ability s . Given prices (w, p_n) , the pair (c_a^W, c_n^W) solves the worker's consumption problem.
- (v) **Markets clear.** For the agricultural and non-agricultural goods, and for land and labor:

$$N_a \int_s c_a^F(s) dF(s) + N_n c_a^W + N_a \int c(x(s, D)) dF(s) = Y_a = N_a \int_s y(s, D) dF(s),$$

$$N_a \int_s c_n^F(s) dF(s) + N_n c_n^W = Y_n = AN_n,$$

$$N_a \int \ell(s, D) dF(s) = L,$$

$$N_a + N_n = 1.$$

4.3 Model Implications

Given climate-adjusted productivity z , the optimal per-period profit of a farm with idiosyncratic productivity s is

$$\pi(sz) = (1 - \gamma) \left(\frac{\gamma}{q} \right)^{\frac{\gamma}{1-\gamma}} (A\kappa)^{\frac{1}{1-\gamma}} sz.$$

For notational convenience we define

$$\Omega \equiv (1 - \gamma) \left(\frac{\gamma}{q} \right)^{\frac{\gamma}{1-\gamma}} > 0,$$

so that

$$\pi(sz) = \Omega (A\kappa)^{\frac{1}{1-\gamma}} sz.$$

Using the assumed climate-adjusted productivity technology:

$$z(D, x) = e^{-D} (1 + x)^{\eta_0 + \eta_1 D}, \quad \eta > 0,$$

and the convex per-period investment cost

$$k(x) = \frac{\psi}{1 + \nu} x^{1+\nu}, \quad \psi > 0, \nu \geq 0,$$

a farm with productivity s and damage D chooses x to maximize

$$\Pi(s, D, x) = \Omega (A\kappa)^{\frac{1}{1-\gamma}} s e^{-D} (1 + x)^{\eta_0 + \eta_1 D} - \frac{\psi}{1 + \nu} x^{1+\nu}, \quad x \geq 0.$$

The first-order condition (FOC) for an interior optimum $x > 0$ is

$$\frac{\partial \Pi}{\partial x} = \Omega (A\kappa)^{\frac{1}{1-\gamma}} s e^{-D} \eta D (1 + x)^{\eta_0 + \eta_1 D - 1} - \psi x^\nu = 0, \quad (12)$$

which can be written as

$$\frac{\Omega (A\kappa)^{\frac{1}{1-\gamma}} s e^{-D} \eta D}{\psi} = \frac{x^\nu}{(1 + x)^{\eta_0 + \eta_1 D - 1}}.$$

5 Quantitative Analysis

We calibrate a benchmark economy to match data moments for Italy. We then perform a series of counterfactual experiments to assess the role of factors that shape the farm size distribution on climate adaptation decisions and their macroeconomic consequences.

5.1 Calibration

We calibrate a benchmark economy to micro, sectoral, and aggregate data for the Italy. We proceed in two steps. We first calibrate to a benchmark economy with no climate damages $D = 0$. In this step we can discipline all the parameters except the four adaptation technology parameters $(\eta_0, \eta_1, \psi, \nu)$. We then assume climate damages $D = 0.1$ and calibrate the remaining four parameters to match the empirical estimates from the Italian microdata.

We assume that farming ability s is drawn from the lognormal distribution with normalized mean 0 and standard deviation σ_s . The model features nine parameters to calibrate: the decreasing returns to land γ , the dispersion in endowed farming ability σ_s , the curvature parameters of climate adaptation benefits and costs (η, ν) , the agricultural consumption weight α , the price elasticity ρ , the income elasticity φ , relative productivity in agriculture κ , economy-wide productivity A , and the aggregate land endowment L .

A set of five parameters are either normalized or disciplined by outside evidence. First, we set the decreasing-returns parameter to $\gamma = 0.54$, consistent with agricultural factor income shares reported by [Valentinyi and Herrendorf \(2008\)](#) and standard in the misallocation literature. We set the price-elasticity parameter to $\rho = 0.40$ and the income-elasticity parameter to $\varphi = 0.70$, following [Hao et al. \(2020\)](#). Aggregate productivity terms are normalized to one, $A = \kappa = 1$.

Step 1. Benchmark economy without climate damages. We calibrate a set of four parameters in the benchmark economy assuming no climate damage ($D = 0$) to data for

Italy in 2000. We calibrate σ_s to match the size distribution of farms in the Italy, the land endowment L to match average farm size in agriculture. We calibrate the preference parameters the level of agricultural employment in the Italian data. See Table 6 for a summary of parameter values and targeted moments.

Table 6: Summary of Calibrated Parameters and Moments

Parameter	Description	Value	Target moment	Model	Data
α	Pref. weight for agriculture	0.22	Ag. employment share	5.0%	5.0%
σ_s	Farming ability dispersion	1.98	Farm size distribution	$\{\dots\}$	$\{\dots\}$
σ_n	Non-ag ability dispersion	0.60	Non-ag income dispersion	0.60	0.60
L	Land endowment	0.38	Average farm size (ha)	7.5	7.5

Step 2. Climate damage and adaptation technology parameters. We now calibrate the parameters governing climate damage and productivity investment in adaptation. We parameterize the climate-adjusted investment as

$$\log z(D, x) = -D + \eta_0 \log(1 + x) + \eta_1 D \log(1 + x), \quad (13)$$

where $D \geq 0$ is a reduced-form damage and $\eta_1 > 0$ captures how strongly adaptation x mitigates damage. Recall that the farming production function implies that farm-level TFP in logs is proportional to $(1 - \gamma) \log z(D, x)$ for a farmer with ability s .

In the empirical implementation, we map D to temperature T via a linear index

$$D(T) = \delta T,$$

so that

$$\log z(D(T), x) = -\delta T + \eta_0 \log(1 + x) + \delta \eta_1 T \log(1 + x),$$

and the climate contribution to log TFP is

$$(1 - \gamma) \log z(D(T), x) = -(1 - \gamma) \delta T + (1 - \gamma) \delta \eta T \log(1 + x). \quad (14)$$

At the micro level, we estimate the reduced-form regression

$$\log \text{TFP}_{it} = \beta_0 + \beta_1 T_{it} + \beta_2 \log(1 + x_{it}) + \beta_3 T_{it} \times \log(1 + x_{it}) + \text{controls} + \varepsilon_{it}, \quad (15)$$

where T_{it} is (demeaned) temperature and x_{it} is the measure of adaptation (e.g. irrigated area). Comparing (14) and (15) with $D(T) = \delta T$ yields the mapping

$$\beta_1 = -(1 - \gamma) \delta, \quad \beta_2 = \eta_0, \quad \beta_3 = (1 - \gamma) \delta \eta_1. \quad (16)$$

Hence, we set

$$\delta = -\frac{\beta_1}{1 - \gamma}, \quad \eta_0 = \beta_2, \quad \eta_1 = -\frac{\beta_3}{\beta_1},$$

so the curvature parameter η_1 is identified by the *ratio* β_3/β_1 and does not depend on γ , while the scale of the damage index δ scales with $(1 - \gamma)^{-1}$.

Our baseline regression estimates $\hat{\beta}_1 = -0.126$, $\hat{\beta}_2 = 0.045$ and $\hat{\beta}_3 = 0.184$, and $\gamma = 0.65$ imply

$$\delta = -\frac{\hat{\beta}_1}{1 - \gamma} = \frac{0.126}{0.35} \approx 0.36, \quad \eta_0 = \hat{\beta}_2 = 0.045, \quad \eta_1 = -\frac{\hat{\beta}_3}{\hat{\beta}_1} = \frac{0.126}{0.184} \approx 0.68.$$

Thus a one-unit increase in T generates a damage index of about $D(T) \approx 0.17T$, and the marginal effectiveness of adaptation increases with damage at rate $\eta_1 \approx 0.68$. Note that the productivity-enhancing investment component of technology that is independent of damages has a relatively small return $\eta_0 = 0.045$.

We next calibrate the two remaining parameters, curvature parameter ν and the scale pa-

parameter ψ of the adaptation cost function, to match moments from our sample over the 2008–2020 period. We assume that the economy features a climate damage shock of $D = 10$ relative.

We discipline the curvature parameter ν using the reduced-form regression of adaptation on farm size and heat shocks,

$$\text{Adapt}_{it} = \gamma_1 \log \text{FarmSize}_i + \gamma_2 \text{HeatShock}_{it} + \gamma_3 (\log \text{FarmSize}_i \times \text{HeatShock}_{it}) + \text{FE} + u_{it}, \quad (17)$$

where Adapt_{it} is (approximately) $\log x_{it}$ and $\log \text{FarmSize}_i$ is the initial farm size. We calibrate cost curvature η to match the elasticity of adaptation intensity with respect to farm size.

Table 7: Summary Calibration Climate Parameters and Moments

Parameter	Description	Value	Source		
<i>Micro estimates</i>					
η_0	Returns to investment	0.045	IV regression estimate		
η_1	Returns to adaptation	0.68	IV regression estimate		
D	Climate damage (BE)	0.10	Estimated damage		
<i>Calibrated under shock $D = 0.10$</i>		Target		Model	Data
ν	Investment cost curvature	2.42	Adaptation–size elasticity	0.09	0.09
ψ	Investment cost scale	1.31	Average adaptation level	0.30	0.30

We choose ψ to match the average level of adaptation observed in the data. Let $\overline{\text{Adapt}}$ denote the sample mean of Adapt_{it} (our measure of $\log(1 + x_{it})$), and let (s_i, D_{it}) denote the farm–damage pairs used in the second-stage regression. Given (δ, η, ν) , we select ψ to solve: $\overline{\text{Adapt}} \approx \frac{1}{N} \sum_{i,t} \log(1 + x(s_i, D_{it}))$.

This targets the average intensity of adaptation, while the curvature parameter ν is disciplined by the cross-sectional response of adaptation to farm size and temperature. Average of $\log(1 + x)$ is around 0.3 in the data.

5.2 Cross-Country Experiments

We generate poorer and richer economies relative to the benchmark economy (calibrated to Italy) by varying economy-wide productivity A and agriculture-specific productivity κ jointly to capture the relationship between aggregate labor productivity and the share of employment in agriculture observed in the data. Specifically, we consider values of (A, κ) ranging from $(0.10, 0.15)$ in the poorest economy to $(1.00, 1.00)$ in the benchmark economy, and to $(2.00, 1.77)$ in the richest economy.

We also pose a simple functional form for the food security threshold to increase with aggregate labor productivity:

$$\bar{a}(Y) = \bar{a}_0 + \bar{a}_1 \left(\frac{Y}{Y_{BE}} \right).$$

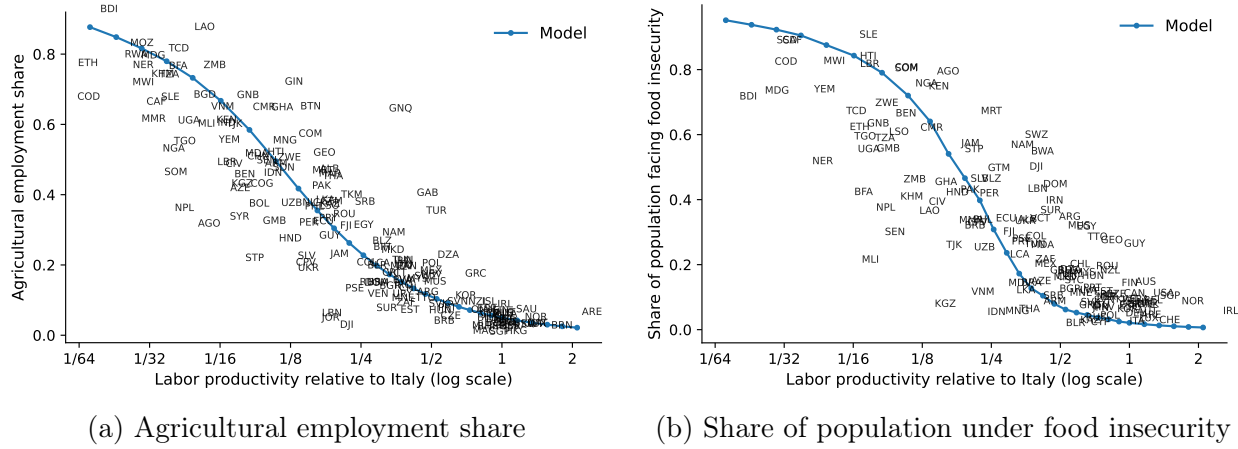
We choose $\bar{a}_0 = 0.05$ and $\bar{a}_1 = 0.05$ to capture the relationship between aggregate labor productivity and food insecurity observed in the cross-country data. Under this calibration, the poorest countries in the sample have a food security threshold of approximately 0.051, while the benchmark economy has a threshold of 0.10. Note that this parameterization of food insecurity only matters for distributional implications and has no implications for real macroeconomic outcomes.

We assume that all other parameter values are the same as in the benchmark economy.

Model fit. Figure 9 compares the model and the data for two key outcomes: the agricultural employment share and the share of the population experiencing food insecurity, both plotted against aggregate labor productivity relative to Italy. The model fits the data very well in both panels. In particular, it captures the cross-country pattern of differences in structural transformation, as well as the relationship between food insecurity and the level of development.

It is worth noting that the model can generate a wide range of development outcomes. Across the range of productivity values we consider, economies span from labor productivity equal

Figure 9: Model vs. Data



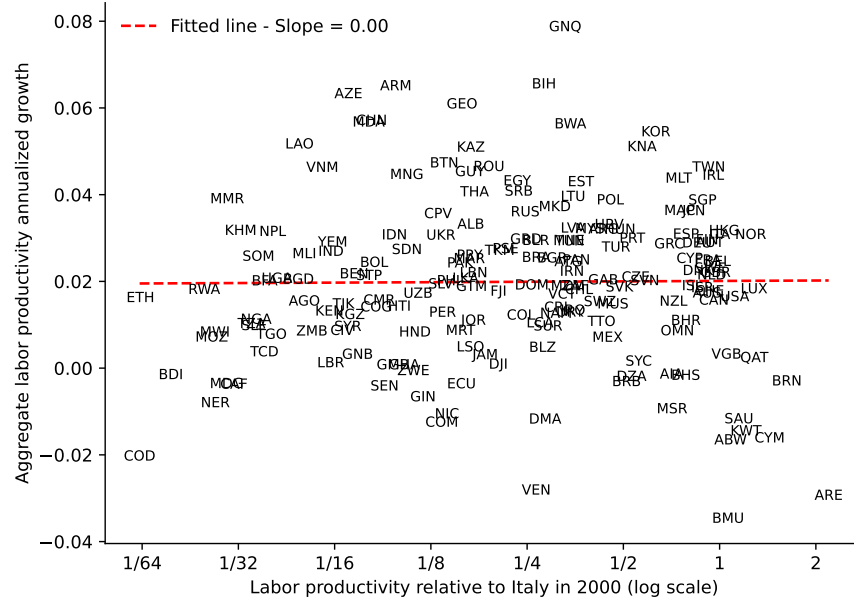
Notes: Panel (a) plots the agricultural employment share against labor productivity relative to Italy in 2000. Panel (b) plots the share of the population experiencing moderate or severe food insecurity against labor productivity relative to Italy. Because the available data are not fully consistent across countries and years, in Panel (b) we normalize labor productivity by Italy's labor productivity in the same reported year. Data on aggregate labor productivity are from PWT 11.0, data on agricultural employment shares are from the International Labour Organization (ILO), and data on food insecurity are from FAOSTAT.

to 1/64 of the benchmark to 2 times the benchmark. Over this range, the model generates agricultural employment shares from about 1% to more than 80%, and food insecurity rates from 0% to more than 80%. Our range of (A, κ) captures most of the observed development spectrum.

Projected growth to 2050. We assume a constant annualized growth rate of economy-wide productivity, g_A , of 2%, together with an additional 1% annual growth rate in agriculture-specific productivity, κ to reflect higher labor productivity growth rate in the agricultural sector, across all economies. This assumption is motivated by the empirical finding that aggregate productivity growth averages around 2% and does not vary systematically with the level of development over the 1990–2018 period.

Given projected growth in 50 years from the baseline in 2000, we report model outcomes for different economies in 2050. We consider three scenarios: (1) no climate damage, (2) a medium-damage scenario in which climate change reduces agricultural productivity by 30%,

Figure 10: Aggregate Labor Productivity Growth across Countries



Notes: Data on aggregate labor productivity are from PWT 11.0. Annualized labor productivity growth rates are estimated by fitting a time trend to log labor productivity over the 1950–2023 period.

and (3) a high-damage scenario in which climate change reduces agricultural productivity by 70%. These scenarios are motivated by our evidence discussed earlier of a wide range of climate effects on potential yields with no systematic differences across countries along the labor productivity distribution.

Table 8 reports how exogenous growth in A and κ affect labor productivity growth, structural transformation, and food insecurity across economies over a 50-year horizon under the zero climate-loss scenario. The table shows a clear catch-up pattern: low-income and middle-income economies grow faster than the high-income benchmark in percentage terms, although substantial aggregate productivity gaps remain. In particular, aggregate labor productivity rises by around 7.7-fold from 0.03 to 0.23 in low-income economies, by around 3.4-fold from 0.19 to 0.65 in middle-income economies, and around 2.8-fold from 1.00 to 2.78 in high-income economies. Low-income economies substantially narrow their aggregate labor-productivity gap to the frontier, with relative productivity rising from 3% to 8%, about a 2.7-fold increase. Middle-income economies also reduce the gap, though more gradually, increasing from 19%

to 23%, about a 1.2-fold increase.

Table 8: Zero Climate Loss Scenario across Growing Economies

	Low income	Middle income	High income
Aggregate labor productivity:			
2000	0.03	0.19	1.00
2050	0.23	0.65	2.78
<i>2050 relative to high-income</i>	<i>0.08</i>	<i>0.23</i>	<i>1.00</i>
Share of population (change in p.p):			
Agricultural employment	−49.0	−19.5	−2.9
Food insecurity	−49.6	−40.3	−1.5

Notes: Low-income economy has $(A, \kappa) = (0.12, 0.18)$; middle-income has $(A, \kappa) = (0.26, 0.33)$; high-income is the benchmark economy with $(A, \kappa) = (1.00, 1.00)$. Outcome variables are computed as changes/growth relative to the case with no growth and no climate damage initially.

The results indicate that aggregate labor productivity catch-up is driven mainly by structural transformation as the exogenous sectoral productivity growth rates are the same across all economies. Low-income economies reallocate labor out of agriculture by 49 percentage points, compared with 19.5 percentage points in middle-income economies and only 2.9 percentage points in high-income economies. This reallocation occurs along with large improvements in food security: the food-insecure population share declines by 49.6 percentage points in low-income economies and 40.3 percentage points in middle-income economies, while the reduction is modest in high-income economies (1.5 percentage points).

Unequal impact of climate damages across economies. Table 9 reports the effects of climate damage relative to each economy’s no-damage growth path, comparing a 30% and a 70% climate-damage scenario. In both cases, climate damage slows structural transformation out of agriculture, with much larger effects in poorer economies. Under 30% damage, the agricultural employment share is 8.26 percentage points higher than in the no-damage path in low-income economies, compared with 1.05 and 0.06 percentage points in middle- and high-income economies. Under 70% damage, these gaps widen sharply to 22.47, 2.35, and

0.19 percentage points, respectively. Food insecurity also rises relative to the no-damage benchmark, again most strongly in poorer economies: the food-insecure share increases by 5.90 and 13.24 percentage points in low-income economies under 30% and 70% damage, versus 1.79 and 3.76 in middle-income economies, and only 0.12 and 0.28 in high-income economies.

Table 9: Impact of Climate Damages Relative to No-Damage Scenario

	30% climate damage			70% climate damage		
	Low-	Middle-	High-	Low-	Middle-	High-
Share of population (change in p.p):						
Agricultural employment	8.26	1.05	0.06	22.47	2.35	0.19
Food insecurity	5.90	1.79	0.12	13.24	3.76	0.28
Agricultural TFP (change in %)	−14.03	−7.52	−2.65	−27.59	−15.43	−8.82
Average productivity losses mitigated by adaptation (%)	<i>17.28</i>	<i>32.52</i>	<i>69.28</i>	<i>14.92</i>	<i>30.33</i>	<i>66.53</i>
Aggregate labor productivity (change in %)	−11.53	−1.38	−0.21	−30.89	−3.05	−0.71
Aggregate labor productivity relative to high-income	0.20 <i>0.07</i>	0.64 <i>0.23</i>	2.78 <i>1.00</i>	0.16 <i>0.06</i>	0.63 <i>0.23</i>	2.76 <i>1.00</i>

Notes: Low-income economy has $(A, \kappa) = (0.12, 0.18)$; middle-income has $(A, \kappa) = (0.26, 0.33)$; high-income is the benchmark economy with $(A, \kappa) = (1.00, 1.00)$. All outcomes are computed as changes relative to the same economy under no climate damage ($D = 0$). Agricultural TFP is computed as $Y_a/(N_a^{1-\gamma}L^\gamma)$. Average productivity losses mitigated by adaptation is computed as the percentage of climate loss mitigated by endogenous investment in adaptation technology on average across farms.

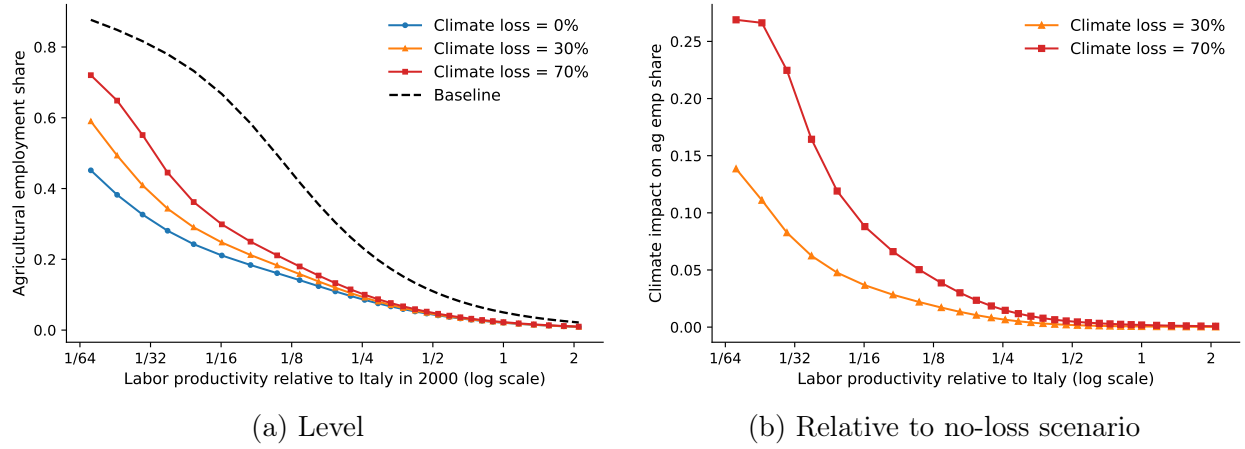
The structural-transformation results are driven by heterogeneous farm-level adaptation investment responses and thereby different impacts on the agricultural TFP between rich and poor countries. Climate damage generates much larger declines in agricultural TFP in poorer economies, with agricultural TFP declines by −14% and −27.6% in low-income economies, compared with −7.5% and −15.4% in middle-income economies, and −2.7% and −8.8% in high-income economies. A key reason is unequal adaptation capacity: low-income economies mitigate only 17% and 15% of average farm-level productivity change, while high-income economies mitigate 69% and 66.5%. Because agriculture remains a large sector in poorer

economies, these agricultural losses translate into much larger aggregate effects, with aggregate labor productivity decline of -12% and -31% in low-income economies, compared with -1.4% and -3% in middle-income economies and -0.2% and -0.7% in high-income economies.

Projected structural transformation patterns in different climate scenarios. Figure 11 reports how agricultural employment responds across economies under alternative climate-damage scenarios, highlighting both levels and changes relative to no climate damage. The figure shows that the model reproduces a strong structural-transformation pattern: technological progress drives labor out of agriculture, but climate damages slow that reallocation even when adaptation is chosen endogenously. Panel 11a shows the agricultural employment share across countries under different climate loss scenarios. In the no-damage scenario, projected productivity growth generates rapid structural transformation over the 50-year horizon, especially for low-income economies. In particular, low-income economies substantially reallocate labor away from agriculture by about 50 percentage points, with agricultural employment shares falling from roughly 80% to about 30%. Middle-income economies also reallocate labor out of agriculture, with agricultural employment shares declining from around 40% to 20%. High-income economies exhibit limited labor reallocation because their agricultural employment shares are already close to long-run low levels.

In panel 11b, we show the impacts of different climate losses relative to the no-damage benchmark across economies. Under the average-loss scenario (around 30% agricultural yield loss), low-income economies retain agricultural employment shares that are about 10–15 percentage points higher than in the no-damage path, indicating slower transformation out of agriculture. Middle-income economies are less affected (generally below 5 percentage points), while high-income economies show close-to-zero effects. Under the high-loss scenario (around 70% yield loss), the slowdown becomes larger: low-income economies are about 15–25 percentage points above the no-damage benchmark, middle-income economies around 5

Figure 11: Projected Structural Transformation across Economies



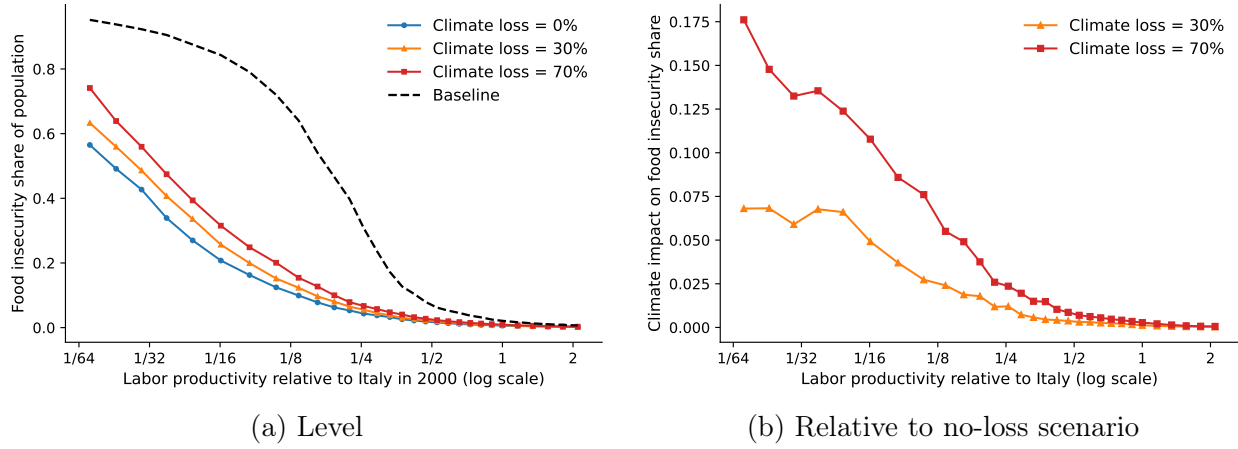
Notes: Panel (a) plots the agricultural employment share against labor productivity relative to Italy in 2000. Panel (b) plots the differences between agricultural employment share in climate damage scenarios vs. the no damage scenario. The x-axis shows the productivity label productivity in 2000.

percentage points, and high-income economies remain only weakly affected.

Projected food insecurity patterns in different climate scenarios. Figure 12 reports how climate damages affect the evolution of food insecurity across countries. The central pattern mirrors the structural-transformation results: technological progress generates large improvements in food security, but climate losses slow these gains, with the strongest effects in low-income economies. Panel 12a shows the share of population facing food insecurity across countries under different climate-loss scenarios. In the no-damage scenario, projected productivity growth over 50 years leads to substantial declines in food insecurity, especially in poorer economies. Low-income economies reduce food insecurity from around 80% of the population to about 40%. Middle-income economies decline from roughly 40% to below 20%. High-income economies start with very low food insecurity shares and exhibit limited additional changes. These patterns indicate that long-run technological progress is a major force improving food security outcomes.

Panel 12b shows the impacts of climate losses relative to the no-damage benchmark. Under the average-loss scenario (around 30% agricultural yield loss), low-income economies have

Figure 12: Projected Food Insecurity across Economies



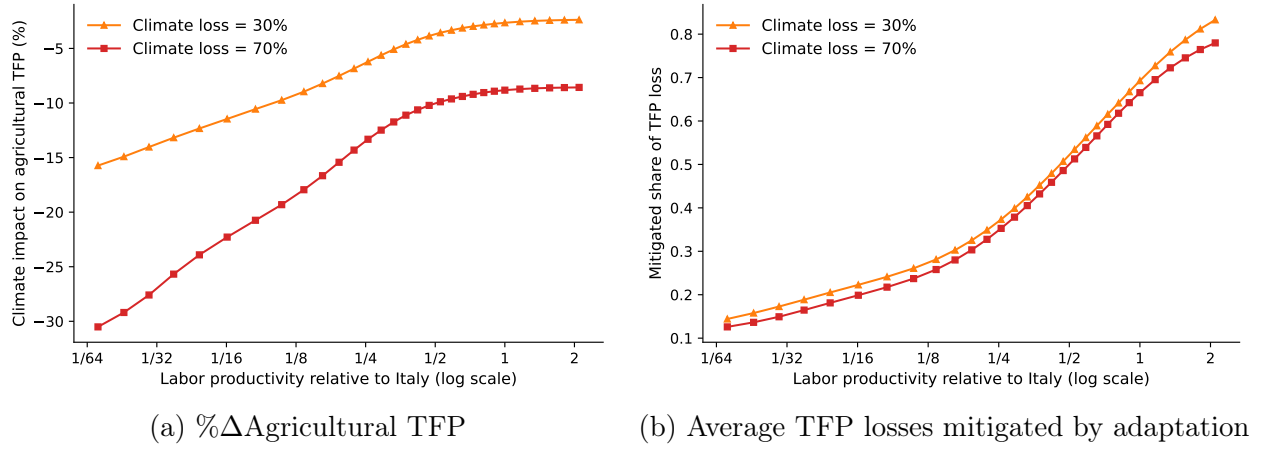
Notes: Panel (a) plots the agricultural employment share against labor productivity relative to Italy in 2000. Panel (b) plots the differences between agricultural employment share in climate damage scenarios vs. the no damage scenario. The x-axis shows the aggregate labor productivity relative to the benchmark economy in 2000.

food insecurity rates about 10–15 percentage points higher than in the no-damage path, while middle-income economies are about 3 percentage points higher. Under the high-loss scenario (around 70% yield loss), the gap widens: low-income economies are about 20–25 percentage points above the no-damage benchmark, and middle-income economies are around 5 percentage points above. High-income economies remain only weakly affected. Overall, climate change materially weakens projected improvements in food security, with the largest burden concentrated in poorer countries.

Climate's impacts on agricultural and aggregate productivity across economies.

Figure 13 extends the table evidence by showing how climate-damage impacts vary across a broader set of countries along the income distribution. The figure confirms the same core pattern: agricultural productivity losses are substantially larger in poorer economies, while richer economies experience smaller losses and offset a larger share through adaptation. Agricultural TFP substantially declines around 10%–15% and 25%–30% for economies at the lowest income levels under 30% and 70% yield loss scenarios respectively. The effects becoming smaller for higher-income economies with the loss of around 5% and 10% under

Figure 13: Impacts of Climate Damages on Agricultural Productivity



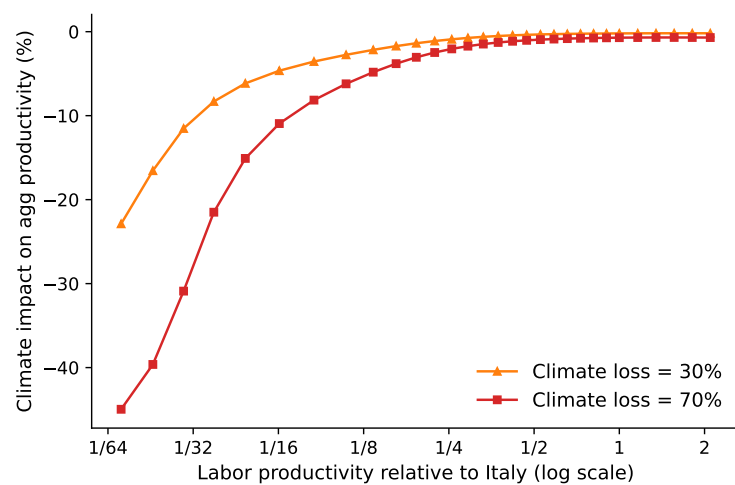
Notes: Agricultural TFP is computed as $Y_a/(N_a^{1-\gamma}L^\gamma)$. %ΔAgricultural TFP is reported as the percentage change in each climate-damage scenario (30% and 70% yield loss) relative to the zero-damage scenario. Average farm-level TFP losses mitigated by adaptation is the average share of climate-induced productivity losses offset by endogenous investment in adaptation technologies across farms.

30% and 70% yield loss scenarios respectively.

While farms in richer economies mitigate a large fraction out to 70%-80% of climate damages through endogenous adaptation investment, farms in lower-income economies mitigate a limited share, only around 15% of the climate losses. Unequal adaptation capacity is a key mechanism behind the larger climate damages on farm-level and sectoral productivity in low-income economies.

Given the large employment share in agriculture and the larger farm-level productivity losses caused by lower adaptation investment, climate change generates much larger aggregate effects in poorer economies. Figure 14 shows how these damages to agricultural productivity translate into economy-wide labor productivity across economies with different income levels. The aggregate effect is close to zero in rich economies under both climate-damage scenarios, while low-income economies face losses of roughly 10%–20% under the 30% yield-loss scenario and up to about 30%–40% under the 70% yield-loss scenario.

Figure 14: Impact of Climate Damages on Aggregate Labor Productivity



Notes: The plot reports the percentage change of aggregate labor productivity in each climate-damage scenario (30% and 70% yield loss) relative to the zero-damage scenario.

6 Conclusion

We studied how climate damages differ across countries at different stages of development through differences in farm structure and adaptation. We showed that low-income countries are characterized by smaller, more subsistence-oriented farms and higher agricultural employment, facing greater exposure to food insecurity. Using farm-level evidence from Italy, we documented that smaller farms invest substantially less in adaptation technologies and consequently experience larger climate-related productivity losses.

We then incorporated this evidence into a structural transformation model with heterogeneous farm sizes, climate-induced declines in agricultural productivity, and endogenous adaptation investment. We calibrate the model to Italian farm-level and aggregate data and use it to quantify the macroeconomic implications of climate damages across countries.

Under plausible climate and growth scenarios, limited adaptation investment slows the reallocation of labor out of agriculture in developing countries, delaying structural transformation. This mechanism also slows convergence in aggregate labor productivity and under a less optimistic growth scenario can lead to widening cross-country income gaps.

Interestingly, in the 50-year growth scenario we find that the climate impacts are relatively small in magnitude relative to the impact of compounded productivity. This suggests that the same policies addressing the food problem by enhancing productivity in the agricultural sector, are effective in helping economies deal with the effects of climate change. Similarly, technological advances in mitigation technologies can also be effective. We have shown a wide range of climate damages at all points of the income distribution, whereas our experiments impose the same climate shock in all economies. Assessing heterogeneous climate shocks can be relevant in addressing the implications for the disparity in macroeconomics outcomes across countries. We leave these important areas of inquiry for future research.

References

- Adamopoulos, T., Brandt, L., Leight, J., and Restuccia, D. (2022). Misallocation, Selection, and Productivity: A Quantitative Analysis with Panel Data from China. *Econometrica*, 90(3):1261–1282.
- Adamopoulos, T. and Restuccia, D. (2014). The Size Distribution of Farms and International Productivity Differences. *American Economic Review*, 104(6):1667–97.
- Adamopoulos, T. and Restuccia, D. (2022). Geography and Agricultural Productivity: Cross-country Evidence from Micro Plot-Level Data. *The Review of Economic Studies*, 89(4):1629–1653.
- Boppart, T. (2014). Structural change and the kaldor facts in a growth model with relative price effects and non-gorman preferences. *Econometrica*, 82(6):2167–2196.
- Burke, M. and Emerick, K. (2016). Adaptation to climate change: Evidence from us agriculture. *American Economic Journal: Economic Policy*, 8(3):106–140.
- Caggese, A., Chiavari, A., Goraya, S. S., and Villegas-Sanchez, C. (2025). Climate change, firms, and aggregate productivity. ECB Working Paper 3084, European Central Bank.
- Chen, C., Restuccia, D., and Santaaulàlia-Llopis, R. (2023). Land misallocation and productivity. *American Economic Journal: Macroeconomics*, 15(2):441–465.
- Cornes, R. C., van der Schrier, G., van den Besselaar, E. J. M., and Jones, P. D. (2018). An ensemble version of the e-obs temperature and precipitation data sets. *Journal of Geophysical Research: Atmospheres*, 123(17):9391–9409.
- Costinot, A., Donaldson, D., and Smith, C. (2016). Evolving comparative advantage and the impact of climate change in agricultural markets: Evidence from 1.7 million fields around the world. *Journal of Political Economy*, 124(1):205–248.
- Deschênes, O. and Greenstone, M. (2007). The economic impacts of climate change: Evidence from agricultural output and random fluctuations in weather. *American Economic Review*, 97(1):354–385.
- Feenstra, R. C., Inklaar, R., and Timmer, M. P. (2015). The next generation of the penn world table. *American Economic Review*, 105(10):3150–3182.
- Fyfe, J., Fox-Kemper, B., Kopp, R., and Garner, G. (2022). Global surface temperature changes in °c relative to 1850–1900.

- Gollin, D., Lagakos, D., and Waugh, M. E. (2014). The agricultural productivity gap. *The Quarterly Journal of Economics*, 129(2):939–993.
- Gollin, D., Parente, S., and Rogerson, R. (2002). The role of agriculture in development. *American economic review*, 92(2):160–164.
- Gollin, D., Parente, S. L., and Rogerson, R. (2007). The food problem and the evolution of international income levels. *Journal of Monetary Economics*, 54(4):1230–1255.
- Hao, T., Sun, R., Tombe, T., and Zhu, X. (2020). The effect of migration policy on growth, structural change, and regional inequality in china. *Journal of Monetary Economics*, 113:112–134.
- Herrendorf, B., Rogerson, R., and Valentinyi, Á. (2014). Growth and structural transformation. In Aghion, P. and Durlauf, S. N., editors, *Handbook of Economic Growth*, volume 2, chapter 6, pages 855–941. Elsevier.
- IPCC (2021). Climate change 2021: The physical science basis. contribution of working group i to the sixth assessment report of the intergovernmental panel on climate change. Technical report, Cambridge, United Kingdom and New York, NY, USA.
- Lagakos, D. and Waugh, M. E. (2013). Selection, agriculture, and cross-country productivity differences. *American Economic Review*, 103(2):948–980.
- Liu, B., Martre, P., Ewert, F., Porter, J. R., Challinor, A. J., Müller, C., Ruane, A. C., Waha, K., Thorburn, P. J., Aggarwal, P. K., Ahmed, M., Balkovič, J., Basso, B., Biernath, C., Bindi, M., Cammarano, D., De Sanctis, G., Dumont, B., Espadafor, M., Eyshi Rezaei, E., Ferrise, R., Garcia-Vila, M., Gayler, S., Gao, Y., Horan, H., Hoogenboom, G., Izaurrealde, R. C., Jones, C. D., Kassie, B. T., Kersebaum, K. C., Klein, C., Koehler, A.-K., Maiorano, A., Minoli, S., Montesino San Martin, M., Naresh Kumar, S., Nendel, C., O’Leary, G. J., Palosuo, T., Priesack, E., Ripoche, D., Rötter, R. P., Semenov, M. A., Stöckle, C., Streck, T., Supit, I., Tao, F., Van der Velde, M., Wallach, D., Wang, E., Webber, H., Wolf, J., Xiao, L., Zhang, Z., Zhao, Z., Zhu, Y., and Asseng, S. (2019). Global wheat production with 1.5 and 2.0°c above pre-industrial warming. *Global Change Biology*, 25(4):1428–1444.
- Nath, I. (2025). Climate change, the food problem, and the challenge of adaptation through sectoral reallocation. *Journal of Political Economy*, 133(6):1705–1756.
- Restuccia, D., Yang, D. T., and Zhu, X. (2008). Agriculture and aggregate productivity: A quantitative cross-country analysis. *Journal of monetary economics*, 55(2):234–250.

- Rohde, R. A. and Hausfather, Z. (2020). The berkeley earth land/ocean temperature record. *Earth System Science Data*, 12(4):3469–3479.
- Rosenzweig, C., Elliott, J., Deryng, D., Ruane, A. C., Müller, C., Arneth, A., and et al. (2014). Assessing agricultural risks of climate change in the 21st century in a global gridded crop model intercomparison. *Nature Climate Change*, 4(9):816–822.
- Valentinyi, A. and Herrendorf, B. (2008). Measuring factor income shares at the sectoral level. *Review of Economic Dynamics*, 11(4):820–835.
- Zhao, C., Liu, B., Piao, S., Wang, X., Lobell, D. B., Huang, Y., Huang, M., Yao, Y., Bassu, S., Ciais, P., Durand, J.-L., Elliott, J., Ewert, F., Janssens, I. A., Li, T., Lin, E., Liu, Q., Martre, P., Müller, C., Peng, S., Peñuelas, J., Ruane, A. C., Wallach, D., Wang, T., Wu, D., Liu, Z., Zhu, Y.-G., Zhu, Z., and Asseng, S. (2017). Temperature increase reduces global yields of major crops in four independent estimates. *Proceedings of the National Academy of Sciences*, 114(35):9326–9331.

Online Appendix

A Descriptives

Table A.1: Summary statistics

Variable	Mean	SD	Median	Min	Max	N
Farm size (hectares)	30.20	45.07	14.92	0.10	944.77	85,423
TFP (log)	0.023	0.789	0.027	-1.808	1.926	85,423
Δ TFP	-0.029	0.641	-0.012	-2.391	2.333	61,007
$\log(\text{Adapt} + 1)$	0.303	0.743	0.000	0.000	4.879	85,422
Temperature anomaly ($^{\circ}\text{C}$)	-0.053	0.779	-0.074	-3.159	2.927	61,007
Labor input (hours)	3,563	2,518	2,800	120	70,800	85,398
Agricultural capital (euro)	36,142	67,316	12,720	0	2,201,184	85,423

Notes: The unit of observation is a farm-year. Farm size (winsorized at 0.5) is measured as operated agricultural land in hectares. TFP denotes farm-level total factor productivity in logs, and Δ TFP is the annual change in log TFP. $\log(\text{Adapt} + 1)$ measures the intensity of drip (micro-) irrigation adoption. Temperature anomaly is defined as the deviation of local growing-season temperature from its long-run mean. Labor input is measured in total annual hours worked. Agricultural capital refers to the value of farm capital stocks.

Table B.2: First-stage regressions: adaptation and temperature interaction

	(1) IV at $t - 1$	(2) IV at t
<i>Panel A: Dependent variable = $\log(\text{Adaptation})$</i>		
IV(Adaptation)	0.283*** (0.054)	0.291*** (0.052)
IV(Adaptation) $\times$ HeatShock	0.015 (0.030)	0.002 (0.031)
HeatShock	-0.225 (0.272)	-0.114 (0.275)
First-stage F (excluded instruments)	17.9	25.6
R^2	0.25	0.27
Observations	3,079	3,270
Year, sector, altitude FE	✓	✓
<i>Panel B: Dependent variable = $\text{HeatShock} \times \log(\text{Adaptation})$</i>		
IV(Adaptation)	0.026 (0.018)	-0.001 (0.030)
IV(Adaptation) $\times$ HeatShock	0.286*** (0.069)	0.281*** (0.056)
HeatShock	-1.158** (0.483)	-1.062*** (0.369)
First-stage F (excluded instruments)	9.6	21.4
R^2	0.44	0.46
Observations	3,079	3,270
Year, sector, altitude FE	✓	✓

Notes: The table reports first-stage regressions corresponding to the IV (2SLS) specifications in Table 5. In Panel A, the dependent variable is $\log(\text{Adaptation}_{it})$; in Panel B, the dependent variable is $\text{HeatShock}_{it} \times \log(\text{Adaptation}_{it})$. The excluded instruments are the log of EU environmental funds received by the farm (“IV(Adaptation)”) and its interaction with the Heat Shock $\text{HeatShock}_{it} \times \text{IV(Adaptation)}_{it}$. Column (1) uses instruments measured at $t-1$; column (2) uses instruments measured at t . All regressions include year, sector, and altitude fixed effects. Standard errors are clustered at the province level. Reported first-stage F -statistics are for the joint significance of the excluded instruments.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (two-sided tests).

B Empirical Findings: Robustness

Table B.3: Robustness: Climate Shocks $\times$ $FarmSize_{i,0}$ Without Main Effect of $FarmSize_{i,0}$

	(1)	(2)
	<i>Dependent variable: log(Drip irrigation + 1)</i>	
Temperature _t	0.005 (0.016)	
Temp _t $\times$ log farm size _{i,0}	0.006*** (0.001)	
Temperature _{t-1}		0.005 (0.018)
Temp _{t-1} $\times$ log farm size _{i,0}		0.006*** (0.001)
Constant	- 0.062 (0.300)	- 0.058 (0.326)
<i>N</i>	85,514	58,780
Year $\times$ <i>SectorFE</i>	✓	✓
Year $\times$ <i>AltitudeFE</i>	✓	✓

Standard errors clustered at the province level in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes: This appendix table reports robustness checks corresponding to the mechanism specification in Table 4. Each column estimates a model in which the effect of climate shocks on adaptation is allowed to vary with farm size, but the main effect of $\log(\text{Farm Size}_{i0})$ is intentionally excluded. The dependent variable is $\log(1 + \text{Drip Irrigation}_{it})$.

Column (1) uses the contemporaneous temperature measure Temp_{it} and its interaction with farm size, while Column (2) uses the lagged temperature measure Temp_{it-1} and its interaction with farm size. As a robustness check, we replaced initial farm size with contemporaneous log farm size; coefficients in both panels remain virtually identical in magnitude and significance.

Notes: This appendix table reports robustness checks corresponding to the mechanism specification in Table 4. Each column estimates a model in which the effect of climate shocks on adaptation is allowed to vary with farm size, but the main effect of $\log(\text{Farm Size}_{i0})$ is intentionally excluded. The dependent variable is $\log(1 + \text{Drip Irrigation}_{it})$.

Column (1) uses the contemporaneous temperature measure Temp_{it} and its interaction with farm size, while Column (2) uses the lagged temperature measure Temp_{it-1} and its interaction with farm size. As a robustness check, we replaced initial farm size with contemporaneous log farm size; coefficients in both panels remain virtually identical in magnitude and significance.

Table B.4: Determinants of Log TFP: Farm Size, Temperature, and Adaptation

	(1)	(2)	(3)
	Log TFP	Log TFP	Log TFP
Log Farm Size	0.0562 (0.107)		0.0411 (0.101)
Temperature	-0.0343** (0.0164)	-0.0217* (0.0124)	-0.0456** (0.0204)
Temperature $\times$ Log Farm Size	0.00387 (0.00585)		0.00842 (0.00552)
Log Adaptation		0.246* (0.125)	-0.444 (0.344)
Temperature $\times$ Log Adaptation		0.000358 (0.00631)	0.0381** (0.0166)
Log Farm Size $\times$ Log Adaptation			0.266** (0.102)
Constant	7.569*** (0.282)	7.604*** (0.227)	7.529*** (0.371)
Observations	99,072	99,071	99,071

All specifications control for year $\times$ sector and year $\times$ altitude fixed effects.

Standard errors clustered at the province level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.5: Climate Adaptation, $TFP_{i,0}$ and Temperature Shocks (Robustness)

	(1)	(2)	(3)
	<i>Dependent variable: log(Drip irrigation + 1)</i>		
$TFP_{i,0}$	0.168*** (0.003)	0.167*** (0.027)	0.106*** (0.034)
Temperature shock		-0.032 *** (0.012)	
$TFP_{i,0} \times$ Temperature shock		-0.006 (0.006)	
Hot days			0.0004 (0.002)
$TFP_{i,0} \times$ Hot days			0.0018** (0.001)
Constant	0.299*** (0.003)	0.295*** (0.039)	0.240*** (0.067)
N	85,514	61,069	85,514
Year $\times$ <i>AltitudeFE</i>	✓	✓	✓
Year $\times$ <i>SectorFE</i>	✓	✓	✓

Robustness: replacing initial farm size with initial farm-level productivity ($TFP_{i,0}$) yields consistent results. Baseline effects remain positive and statistically significant, and the interaction with heat exposure is preserved, supporting the interpretation that more productive farms are more responsive in adapting through drip irrigation. Standard errors clustered at the province level in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.6: Marginal effect of temperature shocks by farm-size quintile

	(1)	(2)
	log(days > 30)	log(days > 35)
Q1 (small)	-0.120** (0.052)	-0.045* (0.024)
Q2	-0.064** (0.027)	-0.020 (0.019)
Q3	-0.069* (0.037)	-0.018** (0.009)
Q4	-0.090 (0.079)	0.008 (0.019)
Q5 (large)	0.004 (0.062)	0.000 (0.025)
Q5 – Q1	0.123* (0.069)	0.045 (0.031)
Year × Sector FE	✓	✓
Year × Altitude FE	✓	✓
Obs.	85,515	73,052

Notes: Entries report average marginal effects of temperature shocks on log(TFP) by farm-size quintile. Standard errors (clustered at the province level) are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

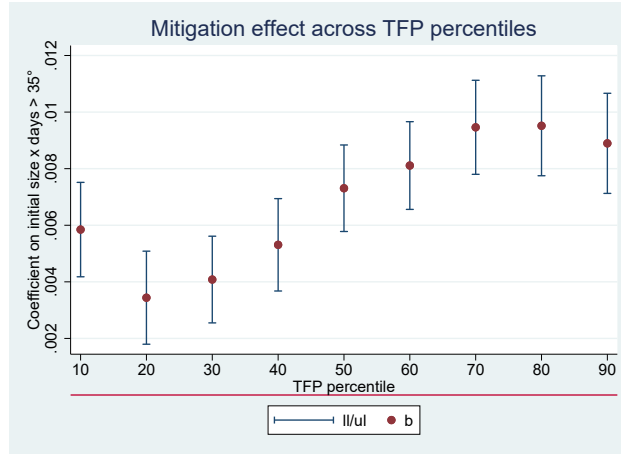


Figure B.1: Quantile regression (deciles): mitigation effect of baseline farm size. Coefficient estimates for the interaction term $\text{avgsz} \times \text{veryhot}$ across deciles of the TFP distribution. A positive coefficient indicates that larger farms experience a weaker productivity loss from very hot days ($> 35^\circ\text{C}$). Vertical bars denote 95% confidence intervals.

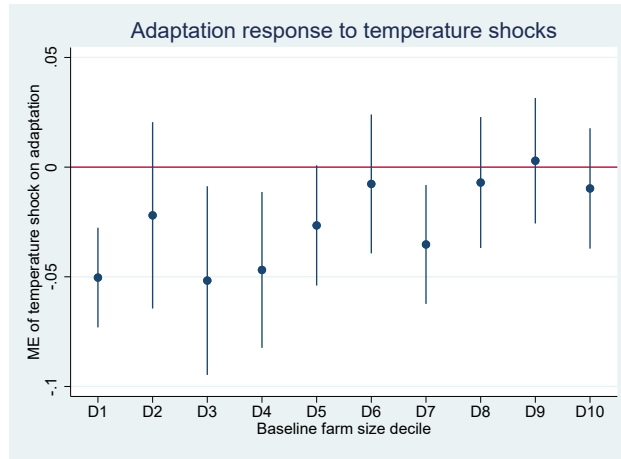


Figure B.2: Adaptation response to Heat Shocks by baseline farm-size decile. The figure reports the marginal effect of Heat Shocks ($dtemp$) on technology adoption, measured as $\log(\text{Drip irrigation} + 1)$, estimated separately across deciles of baseline farm size. The negative response to heat shocks is concentrated among smaller farms (D1–D4) and attenuates substantially for larger farms (D8–D10), where the effect becomes close to zero. Vertical bars denote 95% confidence intervals.

C Other functional forms of climate adaptation $g(D, x)$

We consider several parametric specifications for the damage–adaptation function $g(D, x)$.

(i) Exponential damage with full mitigation. We first consider an exponential form in which adaptation can, in principle, fully offset climate damage. Let $D \in [0, 1)$ denote the productivity loss in the absence of adaptation, so that $g(D, 0) = 1 - D$. We set

$$g(D, x) = 1 - De^{-\eta x}, \quad \eta > 0. \quad (\text{C.1})$$

This specification satisfies

$$g(D, 0) = 1 - D, \quad g(0, x) = 1 \quad \forall x, \quad \lim_{x \rightarrow \infty} g(D, x) = 1,$$

so that a farm with $x = 0$ bears the full damage D , while arbitrarily large adaptation ($x \rightarrow \infty$) restores productivity to its no-damage level. Moreover,

$$g_x(D, x) = D\eta e^{-\eta x} > 0, \quad g_{xx}(D, x) = -D\eta^2 e^{-\eta x} < 0,$$

so adaptation exhibits diminishing returns.

Under the linear profit specification $\pi(sz) = \Omega sz$ derived above and linear cost $c(x) = x$, the first-order condition (??) becomes

$$\Omega s_i D_i \eta e^{-\eta x_i} = 1,$$

which yields the interior optimal adaptation level

$$x(s_i, D_i) = \frac{1}{\eta} \log(\Omega s_i D_i \eta), \quad (\text{C.2})$$

whenever $\Omega s_i D_i \eta > 1$, and $x(s_i, D_i) = 0$ otherwise. This implies that larger and more climate-exposed farms invest more in adaptation.

A convenient feature of (C.1) is that it delivers a simple log-linear representation of $\log g(D, x)$ around the data, which maps directly into reduced-form regressions of log TFP on temperature, adaptation, and their interaction. A limitation is that this specification allows full mitigation: for any given D , sufficiently large x restores $g(D, x)$ arbitrarily close to 1.

(ii) Exponential damage with partial mitigation. To allow for residual climate damages even under very high adaptation, we introduce a parameter $\underline{\phi} \in (0, 1)$ capturing the minimal damage share that cannot be mitigated. We set

$$g(D, x) = 1 - \left[\underline{\phi} + (1 - \underline{\phi})e^{-\eta x} \right] D, \quad 0 < \underline{\phi} < 1, \eta > 0. \quad (\text{C.3})$$

This form satisfies

$$g(D, 0) = 1 - D, \quad g(0, x) = 1 \quad \forall x, \quad \lim_{x \rightarrow \infty} g(D, x) = 1 - \underline{\phi}D,$$

so that only a fraction $1 - \underline{\phi}$ of the damage D can be eliminated, and a residual share $\underline{\phi}D$ remains even as $x \rightarrow \infty$. Adaptation remains increasing and concave:

$$g_x(D, x) = (1 - \underline{\phi})D\eta e^{-\eta x} > 0, \quad g_{xx}(D, x) = -(1 - \underline{\phi})D\eta^2 e^{-\eta x} < 0.$$

The parameter $\underline{\phi}$ has a transparent interpretation,

$$\underline{\phi} = \frac{\text{residual damage under arbitrarily high adaptation}}{\text{damage with no adaptation}},$$

and we therefore treat $\underline{\phi}$ as an externally calibrated parameter, using quantitative estimates of the maximal fraction of climate damages that can be avoided through adaptation in the agricultural sector. Given $\underline{\phi}$, the remaining parameters (δ, η) can be estimated by matching the model-implied response of log TFP to temperature, adaptation, and their interaction to the reduced-form coefficients.

(iii) Hyperbolic damage with partial mitigation. An alternative specification replaces the exponential attenuation with a rational function, which decays more slowly in x :

$$g(D, x) = 1 - D \left[\underline{\phi} + (1 - \underline{\phi}) \frac{1}{1 + \eta x} \right], \quad 0 < \underline{\phi} \leq 1, \eta > 0. \quad (\text{C.4})$$

Again,

$$g(D, 0) = 1 - D, \quad g(0, x) = 1 \quad \forall x, \quad \lim_{x \rightarrow \infty} g(D, x) = 1 - \underline{\phi}D,$$

so that only a fraction $1 - \underline{\phi}$ of damage can be mitigated. The marginal effect of adaptation is

$$g_x(D, x) = D(1 - \underline{\phi}) \frac{\eta}{(1 + \eta x)^2} > 0,$$

with $g_{xx}(D, x) < 0$, implying diminishing returns to adaptation.

Relative to the exponential case, the hyperbolic form implies that mitigation effects taper off more gradually for large x , which can be desirable in counterfactuals that consider large changes in adaptation. Moreover, expanding $\log g(D, x)$ around the sample mean of (D, x) yields a log-linear approximation in temperature, $\log x$, and their interaction, closely aligned with the empirical specification used in our regressions. As in (C.3), the parameter $\underline{\phi}$ can be calibrated from external evidence on residual damages, while η is disciplined by the estimated elasticity of log TFP with respect to adaptation and its interaction with temperature.

(iv) Regression-consistent wedge. To match the reduced-form specification

$$\log \text{TFP}_{it} = \beta_0 + \beta_1 T_{it} + \beta_2 \log x_{it} + \beta_3 (T_{it} \log x_{it}) + \varepsilon_{it},$$

we assume that the climate-adaptation multiplier takes the form

$$z_i = g(T_i, x_i) = \exp(-D(T_i)) x_i^{\psi + \eta D(T_i)},$$

so that

$$\log z_i = -D(T_i) + D(T_i) \eta \log x_i + \psi \log x_i.$$

With the linear damage index $D(T) = \delta T$, this implies

$$\log \text{TFP}_{it} = \alpha_i - \delta T_{it} + \delta \eta T_{it} \log x_{it} + \psi \log x_{it},$$

and therefore

$$\beta_1 = -\delta, \quad \beta_2 = \psi, \quad \beta_3 = \delta \eta.$$

The specification has several advantages and disadvantages.

Pros.

- **Direct link to the reduced form.** With a linear damage index $D(T) = \delta T$, the model reproduces exactly the empirical regression

$$\log \text{TFP}_{it} = \beta_0 + \beta_1 T_{it} + \beta_2 \log x_{it} + \beta_3 (T_{it} \log x_{it}) + \varepsilon_{it},$$

with a one-to-one mapping

$$\beta_1 = -\delta, \quad \beta_2 = \psi, \quad \beta_3 = \delta \eta.$$

Thus the estimated coefficients $(\beta_1, \beta_2, \beta_3)$ can be interpreted directly as structural

parameters (δ, η, ψ) .

- **Transparent interpretation.** The term $-D(T)$ captures the pure effect of temperature on productivity, $D(T) \eta \log x$ captures the fact that adaptation is more valuable in hotter conditions, and $\psi \log x$ allows for a (possibly small) direct productivity effect of adaptation even in the absence of climate damage.
- **Tractability and closed-form decisions.** Under the linear profit specification $\pi(sz) = \Omega sz$ and linear cost $c(x) = x$, the first-order condition for adaptation,

$$\frac{\partial}{\partial x} [\Omega s g(T, x)] = 1,$$

becomes

$$\Omega s \exp(-D(T)) x^{\psi + \eta D(T) - 1} (\psi + \eta D(T)) = 1,$$

which yields a closed-form expression for $x(s, T)$ whenever $\psi + \eta D(T) \neq 1$. This facilitates comparative statics and quantitative counterfactuals.

- **Monotonicity in adaptation.** For $D(T) \geq 0$ and $\psi + \eta D(T) > 0$, we have

$$\frac{\partial \log z}{\partial \log x} = \psi + \eta D(T) > 0,$$

so that an increase in adaptation x always raises climate-adjusted productivity.

Cons.

- **No built-in partial mitigation.** For fixed $T > 0$ and $\psi + \eta D(T) > 0$, we have $z(T, x) \rightarrow \infty$ as $x \rightarrow \infty$. Hence adaptation can more than fully offset climate damage and raise TFP arbitrarily above the no-damage benchmark. This may be at odds with the idea that even very large adaptation efforts cannot completely undo climate change, or that there is an upper bound on feasible productivity.
- **Marginal effect of temperature can change sign at high adaptation.** The marginal effect of temperature on log productivity is

$$\frac{\partial \log z}{\partial T} = -D'(T) + D'(T) \eta \log x = D'(T) (\eta \log x - 1).$$

For moderate x (so that $\eta \log x < 1$) this is negative, as desired. However, for sufficiently large x the model implies $\partial \log z / \partial T > 0$, i.e. additional warming becomes beneficial once adaptation is very high. Empirically, the parameter estimates may ensure that this region is far outside the observed support of x , but the possibility is

present in the functional form.

- **Adaptation at zero damage if $\psi > 0$.** When $D(T) = 0$, we obtain $z(0, x) = x^\psi$. If $\psi > 0$, adaptation has a strictly positive effect on productivity even in the absence of climate damage, and with linear costs the optimal choice $x(0)$ is strictly positive. This violates the normative condition that farms would not invest in climate adaptation when there is no climate risk. One can enforce $x(0) = 0$ by imposing $\psi = 0$ in the structural model and treating a small estimated β_2 as capturing other, unmodelled channels.
- **Unbounded gains in extreme counterfactuals.** Because $z(T, x)$ grows without bound in x for any fixed $T > 0$, aggressive adaptation policies in counterfactual simulations (very large x) can generate unrealistically large productivity gains. If the analysis focuses on moderate changes around the observed range of (T, x) , this may be of limited concern, but for large departures from the data a bounded specification (such as the hyperbolic or exponential partial-mitigation forms discussed above) may be more appropriate.